Industrial Automation and Intra-Household Labor Supply

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Abstract

How does industrial automation reshape the division of labor within the family? While automation disproportionately displaces male-dominated routine tasks and raises women's relative potential wages, I document a reallocation of household time that challenges standard insurance motives. Using American Time Use Survey (ATUS) data and a shift-share instrument for local robot exposure, I find that greater exposure induces women to withdraw from market work in favor of childcare and leisure, while men intensify their market effort. This divergence is driven by a behavioral reversal at the equal-earnings threshold and is amplified in commuting zones with traditional religious environments. To rationalize why these identity-maintenance frictions override the standard household income effect, I estimate a structural household model where breadwinner norms act as a wedge on the wife's effective wage. The model replicates the observed non-monotonic time-use profiles and quantifies the resulting distortion: this identity friction doubles the gender gap in market hours relative to a norm-neutral baseline.

Keywords: Childcare; Gender identity penalties; Household labor supply; Industrial automation; Relative wages.

JEL Codes: J16, J22, J31, D13, O33, F16.

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1 Introduction

Over the last few decades, industrial automation has fundamentally reshaped the labor market landscape in advanced economies. The diffusion of industrial robots disproportionately displaces routine, brawn-intensive tasks—traditionally performed by men—while simultaneously increasing the relative demand for interpersonal tasks, areas where women typically possess the advantage (de Vries et al., 2020). Consequently, this technological adoption functions as a gender-equalizing force, narrowing the gender wage gap by devaluing male-dominated routine labor (Ge and Zhou, 2020; Giuntella et al., 2024), and decreasing the marriage market value of men (Anelli et al., 2021). Figure 1 plots the distribution of occupational robot exposure by sex.[1] Occupations at the upper end of the distribution (i.e., those with high automation risk) are predominantly male, whereas those at the lower end are predominantly female.

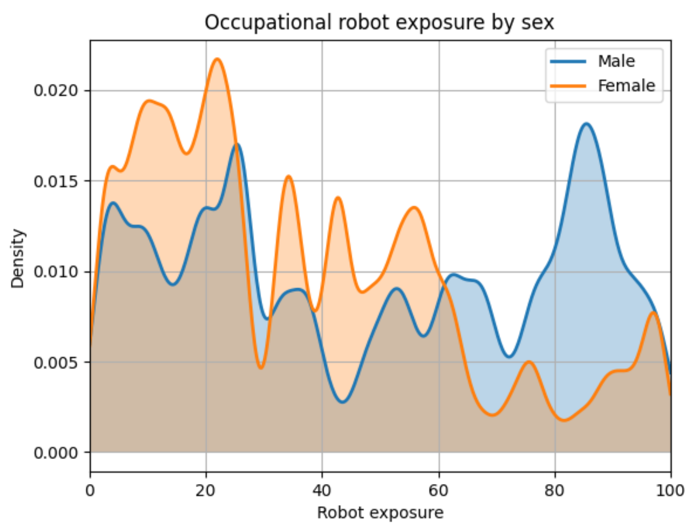


Figure 1: Kernel densities of robot exposure by sex. Source: United States IPUMS Census/ACS microdata, 2000–2020 (Flood et al., 2024). The figure displays the distribution of the robot-exposure score constructed by Webb (2019).

This study traces the consequences of these structural shifts for intra-household time allocation, focusing on how automation, by raising women's relative wages, reshapes the balance between market labor and household production.

Frictionless models of household labor supply predict a clear directional response to such shifts: as a spouse's relative wage and productivity rise, that spouse should substitute time away from home production and toward market work (Becker, 1965; Chiappori, 1992).

[1] Webb (2019) constructs robot exposure by measuring the textual overlap between O*NET task descriptions and robot-related patent text.

Furthermore, because the magnitude of male wage erosion typically strictly dominates any female complementarity gains, industrial automation represents a net negative wealth shock to the household's aggregate expected income. Under standard frictionless models, this overall reduction in household resources should theoretically trigger an added worker effect where the wife increases her market labor to smooth household consumption (Stephens Jr., 2002). Within this framework, industrial automation should unlock female labor supply through both substitution and income channels.

In this paper, I document a behavioral pattern that challenges this standard prediction. I begin by highlighting descriptive evidence from the American Time Use Survey (ATUS) that reveals a non-monotonicity in how couples allocate time. In the cross-section, as a wife's share of household potential wages rises toward parity, her market hours increase; however, when she crosses the equal-wage threshold, this trend reverses. On average, women whose potential hourly earnings surpass their husbands' withdraw from market work to increase childcare and leisure, while their husbands compensate by increasing market effort.

Because cross-sectional correlations are subject to endogenous sorting and unobserved preferences, I utilize local labor market exposure to industrial robots to test whether this pattern represents a causal behavioral response to shifting relative market value. Since automation disproportionately depresses male earnings in the lower-middle segment of the distribution, it acts as an exogenous force pushing a significant mass of households toward the earnings-parity threshold. The households most affected by this shock are those where women's relative potential earnings are already high; for these couples, the technological shock moves them further into a range where the identity-maintenance friction becomes increasingly powerful as the wife's relative wage share rises. By linking ATUS time diaries to a shift-share measure of robot exposure, I find that this shock induces the reallocation suggested by the descriptive evidence: greater exposure causes women to withdraw from market work in favor of home production and leisure by approximately 45 minutes per day per standard deviation of the shock, while men increase their market hours and reduce leisure.

Consistent with an family identity-based mechanism, this behavioral response is nearly three times larger in commuting zones with a high share of traditional religious adherents (e.g., Southern Baptist, Mormon). These results suggest that a significant portion of the documented reallocation is likely driven by intra-household frictions regarding family specialization, where "breadwinner" norms impose a psychological or social penalty as the wife's earnings approach or surpass those of her husband (Bertrand et al., 2015).

To address endogeneity concerns inherent in shift-share measures, I instrument U.S. industry-level robot adoption with contemporaneous adoption trends in five European economies:

Denmark, Finland, France, Italy, and Sweden. This strategy isolates variation in relative wages driven by global technological frontiers, orthogonal to local labor market conditions or regional gender attitudes. The findings are robust to a comprehensive battery of sensitivity checks. I first validate the labor market channel by documenting a significant erosion of male wages and a narrowing of the gender wage gap within the ATUS sample. The results remain robust to adjusting for endogenous sample selection into marriage, utilizing usual weekly hours as an alternative labor supply measure, and accounting for regional trends in the automotive industry. Finally, the estimates remain significant when accounting for the correlation in residuals across areas with similar sectoral compositions, performing a falsification test that isolates the wage-share transmission channel, and controlling for concurrent trade shocks (the "China shock").

To quantify the economic magnitude of these norms, I develop and estimate a structural, semi-cooperative household model of time allocation. The model serves two primary objectives: first, to isolate the proportion of the documented time-reallocation driven specifically by intra-household social norms, and second, to quantify the resulting efficiency losses. Unlike traditional identity models that treat intra-household frictions as separable utility costs, I model the norm as a wedge on the wife's effective market wage, following Greenwood et al. (2016), where social frictions manifest as a reduction in the consumable value of labor. Conceptually, this wedge represents the resource redirection required to maintain the "breadwinner" identity when the traditional household hierarchy is threatened—ranging from compensatory consumption for the husband to increased domestic friction—which effectively reduces the marginal consumable value of the wife's earnings as spouses reach parity.

I estimate the model parameters using the Simulated Method of Moments (SMM) to match the non-monotonic time-use profiles observed in the ATUS. The structural framework provides a conservative lower bound that accounts for approximately 18% of the absolute magnitude of the empirical labor withdrawal while rationalizing the entirety of the directional shift away from the standard added-worker effect. The structural estimates reveal that identity penalties approximately double the gender gap in market hours relative to an identity-neutral benchmark. Counterfactual analyses simulating automation-induced shocks to men's wages highlight two findings: (i) the norm induces a kink at the equal-earnings threshold, generating an inverse-V shape in women's market hours; and (ii) the norm depresses welfare relative to the efficient benchmark by reallocating time away from market work, even when wives possess high market productivity.

The paper makes three distinct contributions to the literature. First, it examines how industrial automation reshapes family dynamics. While Anelli et al. (2021) document that automation reduces marriage rates due to a decline in men's marriage-market value, and

Matysiak et al. (2023); Costanzo (2025) link it to fertility choices, recent economic research has explored other margins of household adjustment. For instance, Giuntella et al. (2024) find that while robot-induced shocks depress wages, households attempt to smooth consumption by increasing their indebtedness and borrowing. This study focuses on intra-household time allocation: how automation alters the joint decision-making of spouses regarding the division of labor between market and non-market activities, particularly when gender identity norms constrain the efficient reallocation of labor.

Second, by exploiting variation in spouses' relative potential wages induced by industrial robots, I address methodological concerns regarding the "identity wedge" hypothesis (Bertrand et al., 2015). Literature suggests that apparent behavioral anomalies at the point of earnings parity may stem from misreporting (Heggeness and Murray-Close, 2019; Rosenberg, 2021) or endogenous labor supply convergence (Zinovyeva and Tverdostup, 2021). By utilizing industrial automation as an exogenous source of wage variation, The paper isolate the causal link between relative earnings and time allocation, circumventing the endogeneity of reported earnings.

Third, I develop and estimate a structural household time-use model that reproduces the non-monotonic relationship between time allocation and spouses' wage shares documented in the descriptive data. Unlike prior structural models focusing on general wage shocks (Blundell et al., 2018) or parental education (Gobbi, 2018), this framework explicitly captures how shifts in relative wage shares—specifically those driven by automation—reshape the trade-offs between market work, childcare, and leisure.

The remainder of the paper is organized as follows. Section 2 describes the datasets and presents stylized facts regarding the non-monotonic relationship between spousal wage shares and household time use. Section 3 details the construction of the shift-share instrument, presents the causal estimates of robot exposure on time allocation, and explores the role of gender norms, and reports robustness analyses. Section 4 develops the structural household model, details its estimation, and provides counterfactual simulations to quantify the economic magnitude of the identity friction. Section 5 concludes.

2 Data, sample, and measurement

The analysis integrates individual-level time diaries from the American Time Use Survey (ATUS) with industry-level data on the operational stock of industrial robots from the International Federation of Robotics (IFR).

Time use. The ATUS provides nationally representative measures of daily time allocation based on 24-hour diaries. Since 2003, the survey has been administered to a subset of households that have completed their final month in the Current Population Survey (CPS), allowing for a rich set of linked demographic and labor market variables. While the cross-sectional nature of the ATUS prevents the tracking of individual transitions over time, the survey captures high-granularity snapshots of daily life that are less susceptible to the recall bias found in stylized labor supply questions.

The primary sample is restricted to married or cohabiting households with at least one employed spouse and at least one child aged 10 or younger. This restriction follows Blundell et al. (2018) to focus on households where the trade-offs between market labor and intensive home production are most binding. Market work corresponds to the ATUS category "work and work-related activities." Childcare time is measured by aggregating "Caring for and helping household children," "Activities related to household children's education," and "Activities related to household children's health."

Table 1 reports descriptive statistics for this sample. The data reveal a demographic with high educational attainment; tertiary degree holders represent over 44% of both men and women. Despite this relative symmetry in human capital, the baseline time-use patterns follow a gendered division of labor. Men average 262 minutes of daily market work compared to 132 minutes for women. This relationship is inverted for home production, where women dedicate an average of 143 minutes to childcare—nearly double the 79 minutes recorded for men.

The high standard deviations relative to the means in Table 1 are characteristic of time-diary data, reflecting the variance between workdays and non-workdays (such as weekends) captured in the 24-hour snapshots. Furthermore, the discrepancy between usual weekly hours (44 for men, 23.5 for women) and diary minutes suggests that the sample includes a significant number of part-time working women and over-employed men. As shown by the comparison with the broader population in Appendix Table A1, the presence of young children significantly compresses leisure time for both spouses, redirecting the household's total time budget toward market work and active caregiving.

Table 1: Descriptive statistics: ATUS analysis sample (households with young children)

	Females			Males		
Variable	Obs	Mean	Std. Dev.	Obs	Mean	Std. Dev.
Leisure (min/day)	24,748	226.80	164.97	22,021	262.00	195.74
Childcare (min/day)	24,748	142.94	147.38	22,021	79.19	113.27
Market work (min/day)	24,748	132.19	215.60	22,021	261.95	277.40
Age	24,748	35.49	6.78	22,021	38.05	7.54
Hourly wage (USD)	7,381	16.45	10.49	7,839	18.68	10.21
Weekly earnings (USD)	14,013	757.96	534.22	16,666	1,086.80	594.43
Usual work hours per week	23,048	23.56	19.61	20,463	44.00	14.24
Less than secondary (%)	24,748	7.85	26.90	22,021	8.72	28.22
Secondary (%)	24,748	19.12	39.33	22,021	22.17	41.54
Some college (%)	24,748	26.54	44.16	22,021	24.83	43.20
Tertiary (%)	24,748	46.48	49.88	22,021	44.28	49.67

Notes: The sample is restricted to married/cohabiting households with at least one employed spouse and at least one child aged $\leq$10. Leisure, childcare, and market work are ATUS diary minutes/day. Hourly wage, weekly earnings, and usual hours are derived from the linked CPS files. Education rows report the share (%) within each attainment category.

Industrial robots. Data on the operational stock of industrial robots are sourced from the International Federation of Robotics (IFR). An industrial robot is defined by the IFR as an "automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes." This definition identifies autonomous physical machines capable of performing motor tasks without a dedicated human operator, distinguishing them from traditional fixed-purpose automated machinery.

The gender-asymmetric impact of automation is driven by the specific physical tasks these manipulators perform. As documented by Ge and Zhou (2020) and Cortes et al. (2024), industrial robots primarily substitute for "brawn-intensive" manual tasks—such as heavy-duty welding, palletizing, and metal machining—occupations where men have traditionally held a comparative advantage and remain overrepresented (Acemoglu and Restrepo, 2020). Conversely, in sectors like electronics and precision equipment, robots are utilized for high-speed assembly and optical inspection. While these robots automate repetitive placement tasks, they frequently complement human labor by shifting workers toward non-routine cognitive roles, including quality control oversight and system maintenance. Because women in the manufacturing sector are disproportionately concentrated in these precision-based or supervisory roles, they are less susceptible to direct displacement. Beyond manufacturing, industrial automation generates significant productivity spillovers in the service and social sectors where women possess a comparative advantage. Concretely, the adoption of automation handles routine administrative and data-processing tasks, which allows workers in healthcare, education, and social services to reallocate time toward high-touch interpersonal interactions and complex problem-solving. By reducing the time required for standardized motor or clerical tasks, robots increase the marginal productivity of the interpersonal and cognitive skills that remain difficult to automate. Consequently, robotization functions as a gender-skewed force for relative potential wages by devaluing the physical routine labor where men prevail while protecting or complementing the cognitive and social oversight roles where women possess a comparative advantage.

The trajectory of robot adoption in the United States is shown in Figure 2. While the initial diffusion of industrial robotics began to rise in the early 1990s, the pace of adoption accelerated sharply after the turn of the millennium, reflecting a fundamental shift in the technological frontier of American manufacturing.

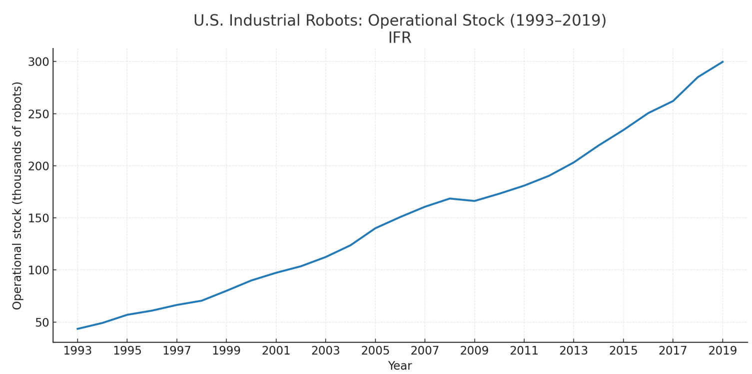


Figure 2: Industrial robot operational stock in the United States, 1993–2019. Data are sourced from the International Federation of Robotics (IFR).

Despite its comprehensive coverage, the IFR dataset presents specific measurement challenges for the United States. Sectoral disaggregation is consistently available only from 2004 onward, as prior years contain a high share of robots not attributed to a specific industry. The analysis leverages detailed data for 13 manufacturing sub-sectors and six non-manufacturing categories, namely: Agriculture and Fishery, Automotive, Construction, Electronics, Food and Beverages, Furniture, Basic Metals, Machinery, Metal Products, Non-metallic Minerals, Mining, Paper, Petrochemicals, Research, Services, Textiles, Utilities, and Non-automotive Vehicles. For robots classified under the "unspecified" category in the original data, I follow the standard literature (Acemoglu and Restrepo, 2020) by allocating them proportionally based on the distribution of robots across the classified industries within the country. Finally, because the IFR provides data only at the national level, I construct a measure of local labor market exposure by mapping these national industry trends to the specific industrial composition of U.S. commuting zones, as detailed in Section 3. This approach accounts for the variation in regional exposure driven by local specialization in robot-intensive industries, effectively transforming national technological shocks into local labor market variation.

2.1 Stylized facts on wage shares and time use

Before estimating the causal impact of automation, this subsection presents descriptive evidence on how spousal wage shares correlate with household time allocation in the cross-section. The data show that when a woman's share of total household wages exceeds her partner's, both spouses reallocate their time between work, childcare, and leisure in distinct, non-monotonic ways. Appendix Section B further shows that these patterns extend to family-oriented beliefs documented in the International Social Survey Programme (ISSP).

Data from the ATUS are used to examine the relationship between spouses' wage shares and household time allocation. The sample is restricted to married or cohabiting couples under age 64 with at least one child aged ten or younger. Following Bertrand et al. (2015), hourly wages are calculated by dividing weekly earnings by weekly hours worked.[2] For non-workers, a potential wage is imputed as the mean wage of employed individuals sharing the same education level, five-year age bracket, state of residence, and survey year. Including non-workers allows for the consideration of adjustments on both the intensive and extensive margins of labor supply. To document these descriptive patterns, the wife's relative status is measured using her share of the hourly wage rate rather than her share of earnings. Relying on actual earnings introduces mechanical endogeneity: because the outcome variables include market hours, women who reduce their working time to comply with the breadwinner norm mechanically lower their actual earnings, sorting themselves into lower bins and obscuring the behavioral threshold. Similarly, using potential earnings in an unconditional cross-section introduces selection bias. Given the baseline gender gap in weekly hours, reaching potential earnings parity requires the wife's hourly wage to be nearly double her husband's. The hourly wage rate share avoids both distortions.

Figure 3 displays the average daily minutes allocated to market work, childcare, and leisure relative to the wife's share of household wages, with separate linear trends fitted on either side of the parity threshold. The inclusion of 95% confidence intervals provides an assessment of the statistical precision underlying these non-monotonic patterns. For women (left column), the narrow shaded regions around the work and childcare estimates show that

[2] Observations implying hourly wages above $100 are dropped due to top-coding truncation.

the slope reversals at the 0.5 threshold are systematic. As the wife's wage share rises toward parity, her working time increases; however, upon crossing the threshold, the trend reverses, forming an inverse-V shape. Her childcare time mirrors this with a V-shaped pattern: it declines as her relative earnings rise toward parity, but reverts to an upward trend once she becomes the primary earner. In contrast, the stable profile of female leisure suggests a countervailing effect between identity frictions and intra-household bargaining. While the breadwinner norm shifts the wife's time toward home production, her rising relative wage share simultaneously strengthens her bargaining position, enabling her to command more private leisure (Chiappori, 1992; Browning and Chiappori, 1998). In the cross-section, these competing channels appear to offset one another, resulting in the observed net-zero effect on female leisure time.

A corresponding reversal characterizes men's time use (right column). Male market work exhibits a V-shape—decreasing as the wife contributes more, but rising once she out-earns him. Conversely, male childcare and leisure follow an inverse-V pattern, increasing up to the parity point before declining. For men, the wider confidence intervals reflect greater idiosyncratic variation in how husbands adjust their time use as they approach and surpass the equal-earnings boundary. These patterns suggest that once the traditional breadwinner hierarchy is threatened, households deviate from standard specialization models, with men increasing market effort and women taking on more non-market tasks to compensate for the norm violation.

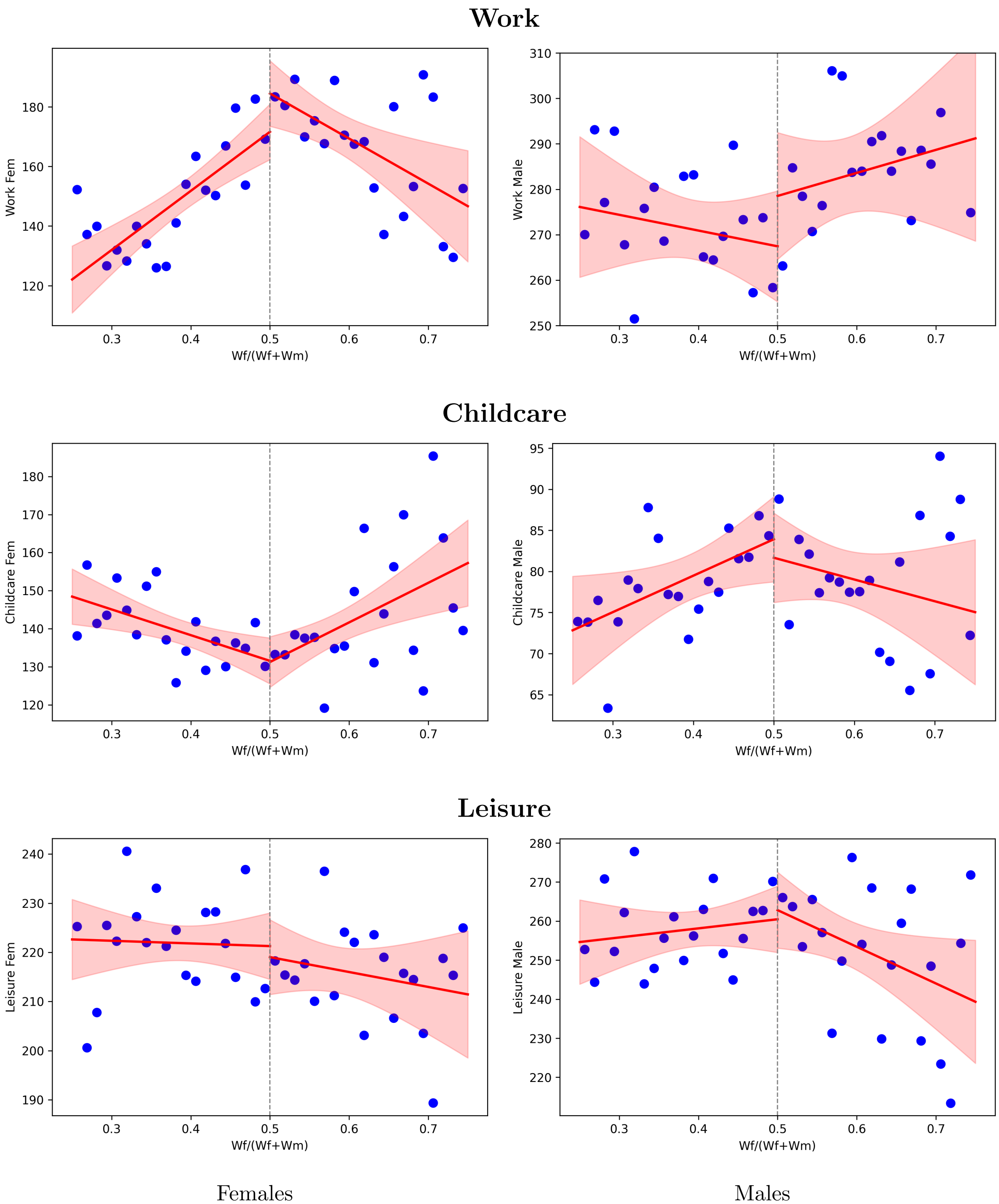


Figure 3: Relationship between a wife's wage share and average daily minutes devoted to work, childcare, and leisure. The figure plots local averages and linear fits allowed to kink at 0.5.

While these descriptive patterns suggest a non-monotonic relationship between spousal wage shares and household labor supply, cross-sectional correlations are inherently endogenous. Spouses may adjust their labor supply in response to unobserved household preferences or local labor market conditions, and women with high career attachment may systematically sort into specific types of marriages. To identify whether this "kink" represents a true behavioral response to shifting relative market value—rather than mere selection—the subsequent empirical analysis (Section 3) utilizes industrial automation as an exogenous shock. Because industrial robots primarily devalue male-dominated routine tasks in the lower-middle segment of the distribution, the resulting wage shocks are concentrated among households where women already hold a high relative wage share. This technological shift moves these families further to the right along the horizontal axis, deeper into the region where the identity norm binds. Consequently, the shock predominantly affects couples already positioned in the area of the distribution where female market work decreases as relative earnings rise, triggering the compensatory reallocation suggested by the stylized facts.

3 The Effect of Robots on Time Use

Motivated by the descriptive non-monotonicities documented in Section 2.1, this section utilizes local exposure to industrial automation to estimate how shifts in relative wages causally affect household time use. The analysis proceeds in three steps. First, I outline the 2SLS empirical framework used to link technological shocks to individual behavior. Second, I describe the construction of the local robot exposure measures and the instrumental variable strategy. Finally, I characterize the distribution of the resulting shocks across U.S. labor markets.

3.1 Empirical Specification

To identify the impact of automation-driven wage shifts on time allocation, I estimate the following baseline specification via two-stage least squares (2SLS):

$$Y_{ict} = \alpha + \beta\,\widehat{\text{Robots}}_{ct} + \eta' C_{ict} + \gamma_c + \delta_{s(c)\times t} + \varepsilon_{ict}, \tag{1}$$

where Y_{ict} represents the daily minutes spent in market work, leisure, or childcare by individual i in commuting zone c and year t. The endogenous regressor, Robots_{ct}, is instrumented in the first stage as:

$$\text{Robots}_{ct} = \pi\,\text{Robots}^{IV}_{ct} + \eta' C_{ict} + \gamma_c + \delta_{s(c)\times t} + u_{ct}, \tag{2}$$

where Robots^{IV}_{ct} is the external shifter defined in Section 3.2. The vector C_{ict} contains a comprehensive set of controls, including temporal factors (day of week, month, holiday), individual demographics (age, education), and household characteristics (number and age of children, spouse's age and education, and family–income category). To account for persistent regional differences and broader economic trends, the model includes commuting-zone fixed effects (γ_c) and state-by-year fixed effects ($\delta_{s(c)\times t}$).

The coefficient β captures the average change in time use for a unit increase in local robot exposure. All regressions are estimated using ATUS survey weights, with standard errors clustered at the commuting-zone–year level. To ensure the results are not driven by cross-regional sectoral shocks, I supplement these baseline estimates with exposure-robust standard errors (Adão et al., 2019) in Section 3.7. Finally, to isolate the structural reallocation of time from cyclical fluctuations, I exclude active job seekers from the sample, thereby reducing bias from direct, short-term unemployment effects.

3.2 Local Exposure and Instrumental Variable

The primary explanatory variable, Robots_{ct}, measures local labor market exposure to industrial robotics using a Bartik-style construction:

$$\text{Robots}_{ct} = \sum_s \frac{\text{Emp}_{sc}^{1990}}{\text{Emp}_c^{1990}} \times \frac{\text{StockRobots}_{st}}{\text{Emp}_s^{1990}}, \tag{3}$$

where $\text{Emp}_{sc}^{1990}/\text{Emp}_c^{1990}$ represents the 1990 industrial specialization of the commuting zone. This approach addresses endogeneity by anchoring regional exposure to historical employment shares before the surge in automation.

However, even with historical shares, U.S. industry-level adoption (StockRobots_{st}) might be correlated with local shocks. To address this, I instrument U.S. exposure using adoption trends in five advanced European economies (EU5). The instrument is defined as:

$$\text{Robots}_{ct}^{IV} = \frac{1}{5}\sum_{j\in\text{EU5}} \left(\sum_s \frac{\text{Emp}_{sc}^{1970}}{\text{Emp}_c^{1970}} \times \frac{\text{StockRobots}_{st}^{j}}{\text{Emp}_s^{j,1990}} \right), \tag{4}$$

where I use 1970 employment shares to further decouple the instrument from recent U.S. economic shifts. Following **?**, identification relies on the quasi-random assignment of global technological "shifts." Validity requires that the subset of industries experiencing rapid global automation is not systematically correlated with unobserved U.S. regional trends in gender norms or labor supply. This allows me to interpret the results as the structural response of households to an exogenous erosion of the husband's comparative advantage.

3.3 Distribution of the Shock

Table 2 provides the summary statistics for the resulting robot exposure measure. While the average exposure in the sample is 3.14 robots per 1,000 workers, this figure masks substantial regional heterogeneity. The intensity of the shock varies by an order of magnitude, from service-oriented hubs like New York City (1.29 robots) to manufacturing centers like Grand Rapids and Detroit, where shocks can exceed 6.03. As shown in the bottom rows of Table 2, this variation is fundamentally driven by historical specialization; the automotive sector's 1990 share is six times higher in the most affected areas than in the least. This ensures that the 2SLS estimates capture the effects of a major structural transformation in the labor market.

Table 2: Distribution and Industrial Profile of the Robot Shock

Statistic	**Value (Robots per 1,000 workers)**
Mean	3.14
Standard Deviation	3.25
10th Percentile (Least Affected)	1.29
50th Percentile (Median)	1.96
90th Percentile (Most Affected)	6.03
99th Percentile (Extreme)	16.65
Maximum	33.70
Mean Automotive Share (1990 Baseline)	
In Least Affected Areas ($< p10$)	1.34%
In Most Affected Areas ($> p90$)	8.25%

Notes: Statistics are calculated at the Commuting Zone-Year level ($N = 1,859$) for 2003–2020. Least affected examples include New York City; most affected examples include Grand Rapids and Detroit.

3.4 Results

This section reports estimation results across three distinct subsamples to isolate the intensive and extensive margins of labor supply. Columns 1 and 4 restrict the sample to households where the spouse works, allowing the respondent's time use to adjust along the extensive margin (including non-employment). Columns 2 and 5 restrict the sample to households where the respondent works, allowing for extensive-margin adjustments by the spouse. Finally, Columns 3 and 6 restrict the sample to dual-earner households, capturing purely intensive-margin adjustments for both partners. In all specifications, Robots_{ct} is instrumented with the external shifter (EU5 industry robot adoption).

Table 3 shows the response of daily time use to a one-unit increase in robot exposure (roughly one additional robot per 1,000 workers).[3]

For men, market work rises significantly: by approximately 14 minutes when allowing for non-employment (column 1), by roughly 8 minutes for respondent-employed individuals (column 2), and by 18 minutes in two-earner households (column 3). This consistent positive response across columns indicates that men attempt to compensate for the automation-induced erosion of their wage rates primarily by increasing their labor effort on the intensive margin. Simultaneously, men's leisure falls by roughly 12 minutes in two-earner households.

For women, the patterns are exactly reversed and remain robust across the various employment subsamples. Market work declines by approximately 15 minutes per day in the full sample (Column 4) and by roughly 19 minutes when conditioning on the respondent's employment (Column 5). In dual-earner households (Column 6), the reduction remains significant at 15 minutes.

The persistence of these negative point estimates across specifications suggests that the female labor supply response operates through both the extensive and intensive margins. While the unconditional decline in Column 4 likely reflects a reduction in labor force participation for some women, the even larger coefficient in Column 5—which is restricted to individuals who remain in the workforce—provides compelling evidence of a substantial intensive margin adjustment. Even among women who maintain their employment status, the robot shock induces a significant scaling back of market hours. This behavior is consistent with a household-level reallocation of time: as the husband's relative labor market position weakens, the wife reduces her market participation, likely by cutting overtime, or transitioning to part-time arrangements.

The observed reduction in market hours for women coincides with an increase in time spent on domestic responsibilities. Specifically, childcare time rises by 6 to 8 minutes across the specifications (Columns 4–6). This shift suggests that the technological shock to the local labor market is associated with a greater concentration of childcare duties among women, even as their market work intensity declines.

The estimates for leisure provide further insight into the potential shift in intra-household dynamics. For men in dual-earner households (Column 3), the 18-minute increase in market work is accompanied by a statistically significant 12-minute reduction in daily leisure. This pattern is consistent with an effort to offset the erosion of hourly wages: to maintain their

[3] In all 2SLS specifications, the Kleibergen-Paap F-statistic for the first stage exceeds the critical values for weak identification (consistently $F > 100$).

absolute earnings contribution to the household, men increase their labor supply by reducing their private consumption of leisure.

The response for women follows a different trajectory, characterized by a significant increase in leisure across all specifications, ranging from 7 to 11 minutes per day. This divergence—where men experience a contraction of leisure while women experience an expansion—may reflect a change in the underlying distribution of household welfare. While gender identity constraints may limit the expansion of female market hours, the simultaneous rise in women's relative potential wages can act as a distribution factor that shifts intra-household bargaining power (Chiappori, 1992; Blundell et al., 2005). In this framework, the increase in leisure suggests that wives are able to capture a larger share of the household's non-market time, even as they absorb a larger portion of the childcare burden.

To contextualize these magnitudes, consider the disparity between the least and most affected regions. In service-oriented hubs like New York City (the bottom decile), where exposure is approximately 1.29 units, the induced reduction in female market work is a relatively modest 19 minutes per day. In contrast, for households in the "automotive heartland" (the top decile) like Detroit or Grand Rapids, where the shock exceeds 6 units, the estimates imply a substantial structural withdrawal of over 90 minutes per day from market work. For the most extreme outliers in the sample (approaching 16 robots per 1,000 workers), the reallocation of time represents a relevant break in the household's daily organization, with women shifting nearly 4 hours per day away from the labor market toward home production and leisure.

Overall, robot exposure drives a distinct, gendered reallocation of time within couples. Rather than women increasing labor supply to offset male wage losses (the standard income effect), women systematically shift out of market work into childcare and leisure, while men intensify their market hours. Because these effects are heavily pronounced even in dual-earner households, they highlight a renegotiation of hours within the marriage rather than simple mechanical displacement.

Table 3: Effect of robots on time spent working, childcaring, and leisuring per day (2SLS Estimates).

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Robots	13.92**	8.376*	18.02**	-15.40***	-18.76***	-15.26**
	(6.598)	(4.618)	(7.384)	(5.073)	(5.899)	(6.628)
Observations	4,100	6,593	3,812	7,194	4,888	4,400
R-squared	0.007	0.003	0.005	0.033	0.006	0.008
R-squared (OLS)	0.426	0.455	0.529	0.297	0.430	0.476
Daily minutes childcare						
Robots	0.160	1.687	-0.773	5.579*	8.220***	8.455**
	(2.533)	(2.111)	(2.707)	(2.975)	(2.701)	(3.383)
Observations	4,100	6,593	3,812	7,155	4,891	4,400
R-squared	0.040	0.044	0.047	0.088	0.066	0.066
R-squared (OLS)	0.170	0.179	0.323	0.230	0.232	0.290
Daily minutes leisure						
Robots	-4.019	-3.869	-11.87***	7.214**	10.13***	10.67***
	(3.400)	(3.365)	(3.818)	(2.968)	(3.186)	(3.918)
Observations	4,100	6,593	3,812	7,194	4,891	4,400
R-squared	0.007	0.006	0.007	0.006	0.010	0.009
R-squared (OLS)	0.251	0.262	0.362	0.215	0.310	0.353
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. Control variables in all specifications include state-by-year fixed effects, spouses' age and education, number of children (total and under age 5), ages of the youngest and oldest child, day-of-week and month-of-survey fixed effects, and a holiday indicator. Standard errors clustered at the commuting-zone–year level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Discussion of Mechanisms and Placebo Evidence. The shifts in time allocation documented in the previous sections suggest a change in the household equilibrium following local automation shocks. Theoretically, these patterns could be driven by several distinct channels. One possibility is a general equilibrium effect, where the displacement of workers in manufacturing reduces aggregate local demand, subsequently depressing female-dominated service sectors. Alternatively, the shock could operate through a regional productivity channel: if automation-driven spillovers increase the marginal productivity of female labor, women might reach a target income level with fewer market hours, leading to a voluntary reduction in labor supply.

To evaluate whether these broader labor market or productivity shocks are the primary drivers, I conduct a placebo test using a sample of individuals without a co-resident partner. These individuals face the same local macro environment, aggregate demand fluctuations, and potential productivity spillovers as those in couples, but they are not subject to intra-household bargaining or identity constraints. As shown in Appendix Table A2, the estimated effects of robot exposure on the market work and leisure of single individuals are close to zero and statistically insignificant. This null result suggests that the main findings are not fundamentally driven by generalized reductions in labor demand or regional shifts in labor productivity, as such shocks would likely manifest across all household types regardless of marital status.

Instead, the evidence points toward mechanisms situated within the intra-household sphere. While multiple factors may be at play, a central candidate is the activation of gender-identity frictions. Because industrial automation disproportionately affects male-dominated occupations, these shocks often push couples toward an earnings distribution that challenges traditional breadwinner norms. If identity-related costs are associated with wives out-earning their husbands, households may respond by scaling back female labor supply to maintain traditional roles. To explore this further, the next section examines the role of local cultural environments. By interacting the robot shock with regional measures of religiosity, I test whether the observed reallocation of time is amplified in areas with more traditional social priors—a result that would confirm intra-household identity dynamics as a primary driver of the results.

3.5 Mechanism: Identity Frictions and the Income Effect

The results presented above suggest a complex interaction between technological change and household labor supply. Since industrial robots primarily displace brawn-intensive tasks—a domain of traditional male comparative advantage—automation theoretically improves the relative market position of women. In a frictionless collective model, a shift in the distribution factor (relative potential wages) alters the intra-household sharing rule. While such models can technically accommodate a range of labor supply responses depending on the preferences of the spouses, the standard economic intuition suggests that an improving relative wage would typically encourage greater market participation. Furthermore, because automation often reduces absolute male earnings, a negative income shock should activate an added worker effect, where women increase their labor supply to maintain household consumption levels (Stephens Jr., 2002). The evidence, however, indicates a simultaneous shift toward reduced market hours for women, a pattern that persists even as their relative economic standing improves.

This divergence from the expected income effect suggests that household responses are moderated by factors beyond simple consumption smoothing. One such factor is the presence of intra-household identity frictions. Exposure to automation disproportionately affects male employment in the lower-middle of the earnings distribution, pushing many couples toward the threshold where the wife's earnings may equal or exceed those of the husband (Bertrand et al., 2015). As the descriptive evidence in Subsection 2.1 suggests, approaching this earnings parity threshold may activate social norms regarding the male breadwinner role. In this context, the psychological or social costs of deviating from traditional gender hierarchies act as an implicit tax on female labor. This friction appears to be a significant determinant of the household equilibrium: rather than increasing labor supply to offset falling male wages, households appear to prioritize the preservation of traditional roles, resulting in a contraction of female market work.

To evaluate this mechanism, I utilize spatial variation in cultural attitudes toward gender roles. I construct a measure of traditional norms using the Churches and Church Membership in the United States, 1990 dataset from the Association of Statisticians of American Religious Bodies (ASARB). By using a 1990 baseline, I ensure that the religious composition of a region is measured prior to the period of intensive robot adoption, reducing concerns that local cultural shifts are endogenous to the economic shocks under study.

Following the literature that associates specific religious affiliations with traditional gender attitudes (Glass and Nath, 2006; Guiso et al., 2003), I calculate the share of the population in each county belonging to denominations that emphasize traditional family structures. These include the Church of Jesus Christ of Latter-day Saints (Mormons), the Southern Baptist Convention, the Assemblies of God, the Lutheran Church–Missouri Synod, and the Presbyterian Church in America, among others. I aggregate these data to the Commuting Zone (CZ) level and normalize by the total CZ population to obtain the local share of traditional adherents.

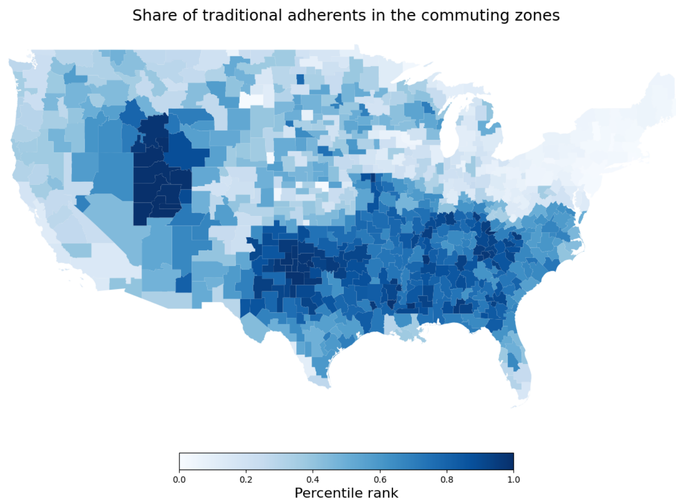


Figure 4: Geographic distribution of traditional religious adherence across Commuting Zones.

Figure 4 illustrates the spatial distribution of this adherence measure. There is a clear concentration of traditional adherence in the Southeastern United States and the Intermountain West. This geographic pattern is distinct from the primary clusters of industrial automation. As previously documented, robot exposure is concentrated in the Manufacturing Belt of the Midwest, driven by the heavy presence of the automotive sector. The limited correlation between these cultural and technological variables is central to the identification strategy; it indicates that the interaction effects reported in Table 4 likely reflect the mediating role of cultural norms rather than the average effects of automation within high-manufacturing regions.

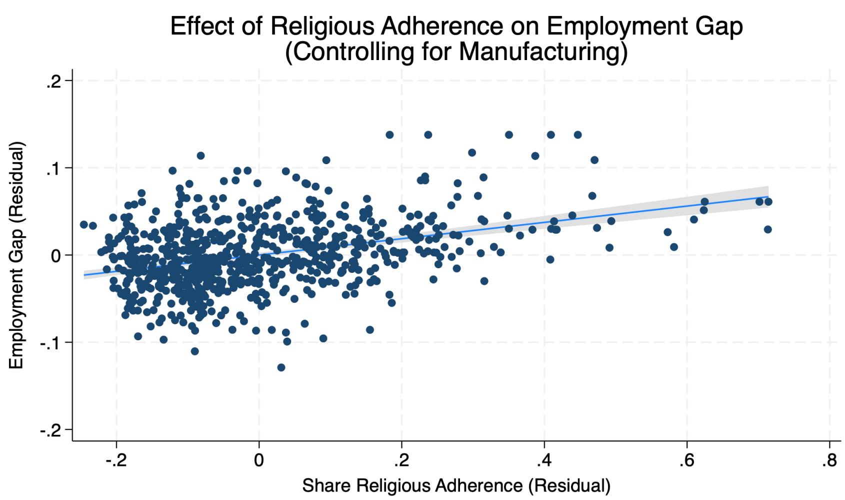


Figure 5: Partial regression of the gender employment gap on traditional religious adherence.

Figure 5 further validates this cultural proxy by showing the relationship between religious adherence and the gender employment gap in 1990. The plot shows the residuals from a regression of both variables on the local share of manufacturing employment. By partialling out the industrial base, the figure demonstrates that even in regions with similar manufacturing intensities, those with higher religious adherence exhibit a wider gender employment gap ($t = 10.85$). This confirms that the adherence measure identifies a persistent cultural dimension of labor supply that is independent of the structural economic factors driving robot adoption.

I define a binary indicator D_c^{High} equal to one if a commuting zone's share of traditional adherents is above the sample median. I estimate an augmented specification that interacts robot exposure with this indicator:

$$Y_{ict} = \alpha + \beta_1 \text{Robots}_{ct} + \beta_2(\text{Robots}_{ct} \times D_c^{High}) + \eta' C_{ict} + \gamma_c + \delta_{s(c)\times t} + \varepsilon_{ict}. \quad (5)$$

In this model, β_1 represents the baseline response in socially progressive regions, while β_2 represents the additional response in conservative regions. Both terms are instrumented using the appropriate interactions of the European robot instrument.

The results in Table 4 reveal a clear gender asymmetry in these interaction effects. For men (Columns 1–3), the interaction term is statistically insignificant for market work, suggests that their labor supply response is driven by economic necessity rather than local cultural priors. In contrast, the results for women (Columns 4–6) show substantial heterogeneity. In progressive commuting zones, an additional robot per 1,000 workers is associated with a 14-minute daily reduction in women's market work (Column 6). In conservative areas, this reduction is amplified by an additional 26 minutes per day. For a household in a high-exposure, conservative environment, the total estimated reduction in female market hours is nearly triple the magnitude of the effect in progressive regions. Because the daily time budget is finite, this amplified withdrawal from the market is accompanied by a significant increase in non-market time. Women in these areas show a more pronounced shift toward domestic responsibilities and a larger increase in residual leisure compared to their counterparts in more progressive regions. This confirms that the technological shock facilitates a return to traditional family specialization most strongly in areas where identity frictions are culturally salient.

Table 4: Heterogeneity by Local Gender Norms (IV Estimates)

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes market work						
Trad × Robots	-5.888	7.377	-2.769	-31.11**	-24.35**	-25.85*
	(15.98)	(14.27)	(15.85)	(12.59)	(11.77)	(15.59)
Robots	12.74**	7.989**	14.01**	-13.16***	-16.74***	-14.41**
	(5.755)	(3.935)	(6.191)	(4.756)	(5.397)	(5.998)
Observations	4,100	6,593	3,812	7,194	4,888	4,400
R-squared	0.007	0.003	0.005	0.032	0.006	0.008
R-squared (OLS)	0.425	0.441	0.477	0.280	0.421	0.430
Daily minutes childcare						
Trad × Robots	5.178	5.730	0.275	19.42**	4.414	-0.176
	(7.328)	(6.103)	(6.883)	(8.603)	(5.580)	(7.253)
Robots	0.496	2.588	1.419	3.075	8.957***	9.350***
	(2.280)	(2.007)	(2.235)	(2.546)	(2.837)	(3.019)
Observations	4,100	6,593	3,812	7,155	4,891	4,400
R-squared	0.040	0.044	0.048	0.088	0.066	0.066
R-squared (OLS)	0.224	0.172	0.246	0.236	0.227	0.241
Daily minutes leisure						
Trad × Robots	-12.72	-11.96	-5.996	6.486	14.36*	21.04*
	(11.39)	(8.519)	(10.76)	(7.169)	(8.718)	(11.01)
Robots	-5.545	-3.493	-8.995***	6.066**	10.22***	8.534**
	(3.830)	(3.116)	(3.196)	(3.031)	(2.918)	(3.482)
Observations	4,100	6,593	3,812	7,194	4,891	4,400
R-squared	0.008	0.007	0.007	0.006	0.010	0.009
R-squared (OLS)	0.288	0.256	0.311	0.206	0.300	0.310
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: The table reports 2SLS estimates of the effect of robot exposure interacted with local gender norms. Trad is a binary indicator equal to one if the commuting zone has an above-median share of adherents to traditional religious denominations in 1990. Sample: married or cohabiting couples with at least one spouse working and children aged 10 or less. Control variables include state-by-year fixed effects, spouses' age and education, number of children, ages of the youngest and oldest child, day-of-week and month-of-survey fixed effects, and a holiday indicator. Standard errors clustered at the commuting-zone–year level are in parentheses. *** p¡0.01, ** p¡0.05, * p¡0.1.

The evidence presented in Table 4 demonstrates that the household response to automation is conditional on local cultural environments. This heterogeneity supports the interpretation that the contraction of female labor supply serves as a functional response to preserve traditional gender hierarchies when technological shocks threaten the male breadwinner status. However, the salience of these identity-related frictions should also vary according to the initial economic distance between the spouses. If the withdrawal of female labor is intended to prevent the wife from out-earning her husband, this behavioral response should be concentrated in households where the spouses are close to earnings parity prior to the shock. The following subsection evaluates this implication by estimating the effects of robot exposure across the distribution of relative spousal wages.

3.6 The Effect of Robot Exposure across the Relative Wage Distribution

The previous analysis demonstrates that the impact of automation is most pronounced in regions where traditional gender norms are culturally salient. To further isolate the role of identity frictions, it is necessary to examine how these effects vary with the baseline economic standing of the couple. In a standard framework of household labor supply, the response to a wage shock depends on the interplay between income and substitution effects. While a collective model of the household is theoretically flexible enough to produce a variety of outcomes—including a reduction in female labor supply if, for example, a shift in bargaining power allows the wife to realize a preference for more leisure—the specific non-linear patterns documented in this section point toward the activation of a discrete gender-identity constraint as couples approach earnings parity.

A significant empirical challenge in evaluating household responses along the relative-earnings distribution is the endogeneity of contemporaneous wages. Relying on realized earnings introduces a mechanical sorting bias: if a woman reduces her labor supply to prioritize childcare following a negative shock to her husband's sector, her actual earnings fall. This behavioral response would mechanically reassign her to a lower relative-earnings position in the data, thereby obscuring the structural dynamics of the household. Furthermore, utilizing contemporaneous potential wages is problematic, as the localized wage structure may have already adapted endogenously to the presence of industrial automation and other regional economic shifts.

To circumvent these endogeneity concerns, I construct a historical, predetermined measure of potential hourly wages using 1990 U.S. Census microdata (IPUMS), a period that predates the primary wave of industrial automation. I calculate the baseline hourly market value of labor by dividing total annual wage and salary income by the product of weeks worked and usual hours worked per week, isolating individuals with positive, valid imputed hourly wages. I then aggregate these data to construct weighted average potential hourly wages within granular demographic cells. These cells are defined by the intersection of four characteristics: sex, state of residence, educational attainment (Less than High School, High School, Some College, and College or higher), and five-year age brackets.

This 1990 historical mapping is subsequently merged into the contemporaneous ATUS sample based on the respondents' and their spouses' current demographic profiles. By fixing the wage-setting structure to 1990 levels, the resulting potential wage share, $s_{pot,i} = w_{f,i}/(w_{f,i} + w_{m,i})$, isolates the household's structural relative price of time. This provides a predetermined measure of the wife's relative status that is orthogonal to both the couple's contemporaneous labor supply decisions and the localized, automation-induced wage fluctuations of the current period. This approach assumes that the relative ranking of demographic groups in the wage distribution remains sufficiently stable over time to identify the structural "threat zone" where earnings parity is approached.

Descriptive Statistics of the Historical Potential Wage Share. Table 5 reports the summary statistics for the assigned 1990 potential hourly wages within the ATUS analysis sample. Because these values are anchored to the demographic and industrial structure of 1990, they reflect the gender wage gap of that era while removing the mechanical penalty associated with historical differences in female labor force participation. The mean absolute potential hourly wage for men is approximately \$12.64, compared to \$8.91 for women (in 1990 dollars), illustrating the structural disadvantage embedded in the historical baseline.

Table 5: Descriptive statistics: 1990 Potential Hourly Wages in the ATUS Sample

Variable	Obs	Mean	Std. Dev.	p10	Median	p90
Female Potential Hourly Wage (\$)	21,522	8.91	1.90	6.40	8.52	11.58
Male Potential Hourly Wage (\$)	20,969	12.64	2.97	8.78	12.23	16.90
Potential Wage Share (s_{pot})	20,443	0.416	0.042	0.365	0.415	0.468

Notes: Sample restricted to married or cohabiting dual-headed households with at least one child aged 10 or younger. Potential hourly wages are constructed as the weighted mean of positive implied hourly earnings from the 1990 IPUMS Census within demographic cells defined by sex, state, four educational categories, and five-year age brackets. Extreme outliers ($<$ \$1 or $>$ \$100 per hour) are trimmed. The potential wage share is defined as $s_{pot} = w_f/(w_f + w_m)$.

The historical wage structure directly influences the distribution of the potential hourly wage share. As shown in Table 5, the mean female potential wage share is 0.416, with a concentrated interquartile range from 0.390 to 0.441. Figure 6 visualizes this distribution, highlighting a structural scarcity of households at the extreme upper tail. Because women's educational attainment and labor market experience have converged with men's since 1990, applying a 1990 wage structure to contemporaneous couples systematically shifts the distribution to the left. Consequently, only 3.32% of the sample explicitly crosses the 0.50 earnings-parity threshold (indicated by the dashed line) under these historical rules.

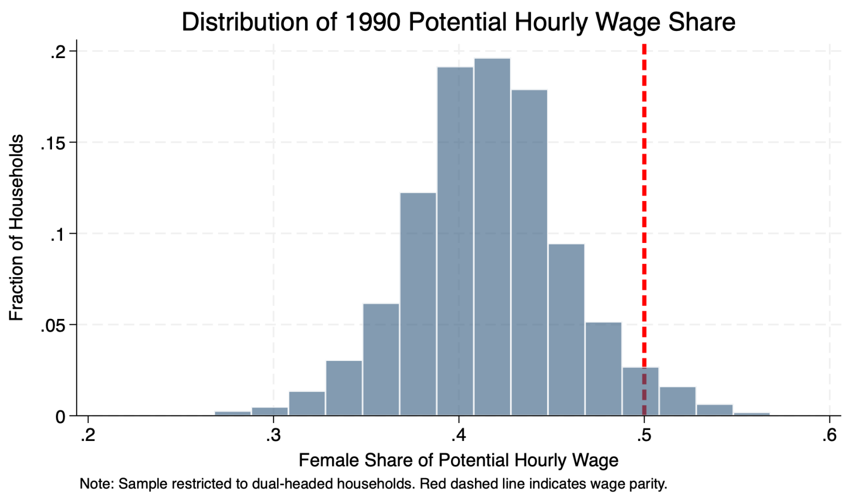


Figure 6: Distribution of the 1990 Potential Hourly Wage Share (s_{pot}) applied to the contemporaneous ATUS sample. The red dashed line denotes the 0.50 earnings parity threshold.

From an identification standpoint, this right-skewness implies that a household assigned a historical potential wage share of approximately 0.45 or 0.47 is likely positioned at or above the true earnings parity threshold in contemporaneous terms. Therefore, rather than relying on a discrete threshold indicator that might artificially restrict the analysis to demographic outliers, the empirical strategy leverages the full continuous distribution of s_{pot}. This allows the model to capture the escalating behavioral responses—the identity friction—as automation pushes couples closer to the modern breadwinner boundary.

Empirical Specification: Continuous Linear Interaction. To capture how the marginal effect of industrial automation evolves as women approach the breadwinner threshold, I estimate a continuous linear interaction model. The sample is trimmed to households where the potential wage share lies in the interval $s_{pot} \in [0.2, 0.6]$, focusing the estimation on the relevant distribution where gender norms are most likely to bind.

I estimate the following two-stage least squares (2SLS) specification:

$$Y_{ict} = \alpha + \beta_1 \widehat{\text{Robots}}_{ct} + \beta_2 s_{pot,i} + \beta_3 (\widehat{\text{Robots}}_{ct} \times s_{pot,i}) + \lambda \ln(w_{m,i}^{pot}) + \eta' C_{ict} + \gamma_c + \delta_{s(c)\times t} + \varepsilon_{ict}, \quad (6)$$

where both the main effect of local exposure (Robots_{ct}) and the interaction term are treated as endogenous and instrumented using the corresponding European adoption shift-share and its interaction with $s_{pot,i}$. The specification is estimated across three subsamples: households where the spouse is employed, households where the respondent is employed, and dual-earner households.

This specification includes the husband's absolute potential wage, $\ln(w_{m,i}^{pot})$. Because industrial automation predominantly displaces routine, lower-wage male labor, the shock falls heaviest on men at specific points in the earnings distribution. Consequently, households with a high female potential wage share are often the same households facing high automation risk by construction of the denominator. If the absolute wage were omitted, the interaction term β_3 might capture a general low-income effect rather than a response to relative gender norms. By controlling for the husband's absolute potential wage, the coefficient β_3 isolates the effect of the relative intra-household earning power. This ensures that the estimated slope is driven by shifts in comparative advantage and the activation of identity frictions rather than a mechanical artifact of the shock's distribution.

Results by Potential Wage Share. Figures 7, 8, and 9 plot the marginal effect of robot exposure on female time allocation across the distribution of the potential wage share. The estimates indicate that the aggregate withdrawal from the labor market is fundamentally driven by households approaching and crossing the earnings-parity threshold, which is consistent with the presence of a persistent identity-maintenance friction.

As shown in Figure 7, when the wife's potential wage share is low ($s_{pot} = 0.2$), the impact of automation on her labor supply is statistically indistinguishable from zero across the various subsamples. In these households, the standard income effect likely mitigates the shock, preventing a sharp drop in female market hours. However, as the potential wage share rises, the marginal effect of automation becomes increasingly negative. At $s_{pot} = 0.6$, the negative effect is large and statistically significant, driving a reduction in market work of roughly 28 to 30 minutes per day per unit of exposure in households where the woman is employed. This downward trajectory suggests that the identity penalty acts as an endogenous marginal tax on female labor that escalates as the breadwinner norm is threatened.

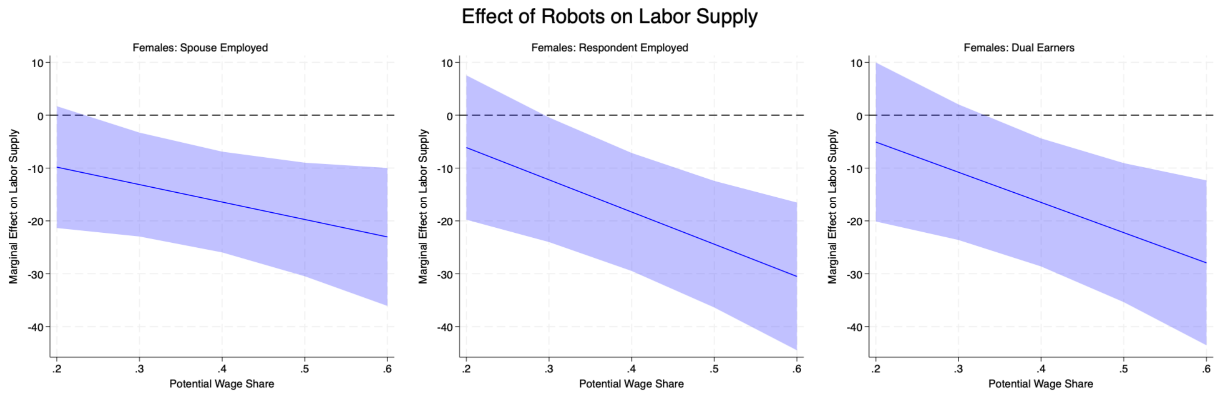


Figure 7: Marginal effect of robot exposure on female daily minutes of market work across the continuous potential wage share distribution (s_{pot}). Shaded areas represent 95% confidence intervals.

The relationship between this market time and home production depends on the wife's baseline employment status. Figure 8 illustrates the effect on childcare. In households where the respondent is already employed or in dual-earner arrangements, the positive effect on childcare is most evident at lower wage shares and attenuates slightly as she approaches parity. Conversely, in the broader sample where only the spouse's employment is guaranteed (Case 1), the childcare effect shows an upward trajectory, becoming significant precisely as the wife approaches parity.

This divergence is consistent with a specialization mechanism. Women in Case 1, who may have greater flexibility on the extensive margin of employment, can specialize in non-market production to compensate for the potential norm violation. This aligns with the heterogeneity results in Table 4, where the increase in childcare time was concentrated in the broader sample, suggesting that domestic specialization is a primary mechanism for identity maintenance in traditional contexts.

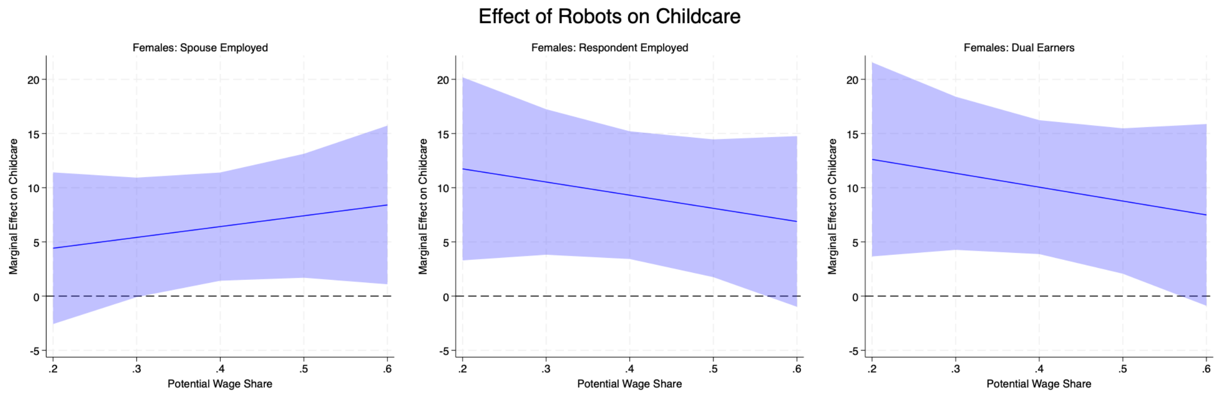


Figure 8: Marginal effect of robot exposure on female daily minutes of childcare across the continuous potential wage share distribution (s_{pot}). Shaded areas represent 95% confidence intervals.

Finally, Figure 9 plots the response of female leisure time, which shows an upward-sloping trajectory across all subsamples. While part of this increase may absorb time displaced from market work, the intensity of the effect also reflects shifts in intra-household bargaining. In collective models, leisure is treated as a private good whose distribution is governed by the bargaining power of the spouses, which is sensitive to the wage ratio (Browning and Chiappori, 1998; Blundell et al., 2005). As automation depresses the husband's relative market value, the wife's structural potential wage share rises, which can increase her relative influence over the allocation of household resources. Even if identity frictions lead her to limit her actual market hours, her underlying earning potential can still enhance her bargaining position. Consequently, women with higher relative wage shares may use this enhanced position to extract a larger share of household time in the form of leisure.

This dual mechanism provides an account of the observed leisure profile: identity frictions act as a constraint forcing time out of the market, while shifts in bargaining power direct that time toward private consumption rather than public household goods. This explains why the marginal effect on female leisure increases from being insignificant near $s_{pot} = 0.2$ to a significant increase of nearly 20 minutes per day near the parity threshold. Ultimately, when technological changes push couples toward traditional gender boundaries, intra-household frictions and bargaining shifts appear to interact and influence the allocation of labor.

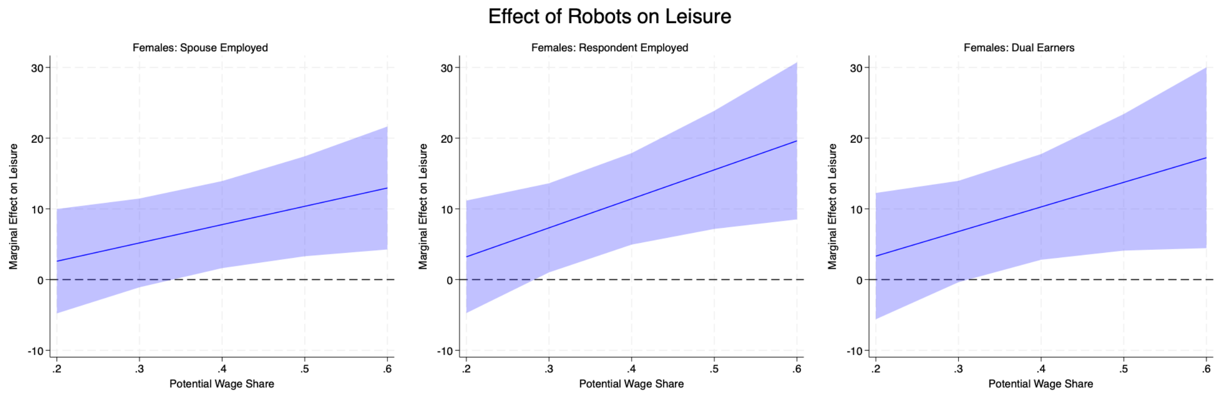


Figure 9: Marginal effect of robot exposure on female daily minutes of leisure across the continuous potential wage share distribution (s_{pot}). Shaded areas represent 95% confidence intervals.

Taken together, these estimates provide evidence that the observed withdrawal of female labor is driven by dynamics situated near the earnings-parity threshold. The negative gradient in market hours as the potential wage share increases suggest that the breadwinner norm can act as an escalating constraint. The reallocation of this time further illustrates the interplay between identity and bargaining. As automation raises the wife's relative status, her increased Pareto weight is associated with a rise in private leisure consumption. Meanwhile, the childcare response indicates how the capacity for domestic specialization varies by employment status. When technological displacement pushes couples against traditional gender boundaries, these intra-household dynamics can lead to distortions in the allocation of labor supply.

3.7 Robustness

The following robustness checks address potential threats to identification and sample validity that may arise in the primary specification, ensuring that the results are not driven by sample selection, idiosyncratic industry trends, or spatial correlation in the residuals.

Representativeness of the ATUS sample: The effect on wages. A potential concern in utilizing the American Time Use Survey (ATUS) to study technology shocks is its smaller sample size and specific coverage compared to the Decennial Census or the American Community Survey (ACS). If the labor market effects of robotization estimated within the ATUS sample were to deviate from established findings, it would suggest that the results on time allocation might be driven by sample selection specificities rather than general economic adjustments. To validate the representativeness of the sample, I estimate Commuting Zone (CZ) level wage regressions using aggregated ATUS microdata.

I employ a stacked long-difference specification rather than the standard annual two-way fixed effects approach used in the main time-use analysis. This choice is motivated by two factors. First, consistent with the automation literature, local labor markets adjust to structural shocks over the medium-to-long run; annual wage data is often subject to transitory fluctuations and business cycle noise that can obscure the structural impact of robot adoption (Acemoglu and Restrepo, 2020). Second, when disaggregated to the CZ level, annual ATUS observations often yield cells with few respondents, leading to substantial measurement error. Using short-term differences would exacerbate attenuation bias due to high sampling error. Relying on stacked differences over 3, 5, and 7 years smooths out high-frequency sampling noise and isolates the structural variation necessary to identify the causal effects of the shocks.

The results in Table A3 indicate that the ATUS sample captures the same wage-rate erosion documented in Census-based studies. In the short run (3-year difference, Columns 1–3), I observe a statistically significant and economically significant negative effect of robotization on male wages ($\beta = -0.061$). This decline drives a narrowing of the gender wage gap ($\beta = -0.050$). Extending the analysis to 5- and 7-year differences, the negative impact on male wages remains robust, although the magnitude of the coefficient attenuates over longer horizons. For a male worker in a top-decile exposure area, these estimates imply a real wage erosion of nearly 36%. This substantial hit to the primary earner's potential income provides the empirical foundation for an income effect. The fact that wives in these households reduce rather than increase their labor supply suggests that the identity-maintenance friction acts as a structural barrier that remains active even under financial necessity.

Correction for sample selection bias. An additional threat to identification arises from endogenous sample selection into marriage or cohabitation. Empirical evidence indicates that negative labor market shocks driven by automation significantly alter family formation dynamics, as the decline in men's marriage-market value in exposed regions can reduce marriage rates and elevate divorce risk (Anelli et al., 2021). Consequently, limiting the analysis to couples with young children may introduce bias if the shocks systematically alter the unobserved composition of households selected into the sample.

To address this, I implement an Inverse Probability Weighting (IPW) procedure. I define the underlying at-risk population as all individuals in the ATUS aged 20–55 and estimate a probit model to predict the probability, $P(S_{ict} = 1)$, that an individual satisfies the baseline sample inclusion criteria. The selection equation is specified as:

$$P(S_{ict} = 1 \mid \text{Robots}_{ct}^{IV}, \mathbf{X}_{ict}) = \Phi\left(\alpha + \gamma \text{Robots}_{ct}^{IV} + \mathbf{X}'_{ict}\beta + \delta_r\right), \tag{7}$$

where $\mathbf{X}_{ict}$ represents a vector of demographic controls and δ_r denotes Census-region fixed effects. I include the external instrument Robots_{ct}^{IV} (European robot adoption) rather than the endogenous exposure measure. Using the instrument in the selection equation ensures that the correction isolates selection driven by exogenous global technological trends rather than by local unobservables that might simultaneously affect adoption and household stability. From the estimated parameters, I calculate the propensity score and construct inverse probability weights $w_{ict}^{IPW} = 1/\hat{p}_{ict}$. Table A4 presents the estimates corrected for selection. The point estimates for market work and leisure are statistically aligned with the baseline results, which suggests that the observed reallocation of time is driven by intra-household adjustments rather than compositional shifts in the household population.

Weekly hours worked. ATUS diaries capture a single day and feature zero-work diaries for individuals sampled on a day off. This raises outcome noise and makes estimates sensitive to functional-form assumptions in the presence of corner solutions. To verify that the findings are not an artifact of this one-day snapshot, Table A5 re-estimates the baseline using respondents' usual weekly hours at their main job from the preceding CPS interview. Because usual hours are collected for workers, these regressions condition on employment. The coefficients remain aligned with the diary-based estimates. For women, a one-unit increase in robot exposure reduces usual working time by approximately 1.0 to 1.3 hours per week (Columns 4–6). This magnitude is consistent with the daily reduction of 15–18 minutes found in the diary estimates, which translates to a similar total over a workweek.

Accounting for the automotive industry. Shift-share estimators can be sensitive to a small number of industries that receive disproportionate weight. If a single industry dominates the identifying variation, the asymptotic properties of the estimator may be compromised. To diagnose this, I compute the Rotemberg weights for the baseline Bartik measure. Table A6 shows that the vehicle sector receives a large weight, reflecting its status as a primary adopter of industrial robots. Because the automotive industry constitutes a concentrated shock, it is necessary to verify that the results are not driven by unobserved regional trends specific to auto-manufacturing hubs. To address this, Table A7 controls for commuting-zone-specific trends across quartiles of employment share in the vehicle sector. The point estimates for both men and women across these specifications remain stable, suggesting the intra-household dynamics are a general response to automation rather than an industry-specific outlier.

Shock-based standard errors (AKM). Consistent with the identifying assumption of exogenous industry-level shifts, standard clustering at the commuting-zone level may understate standard errors if residuals are correlated across regions through shared sectoral shocks. Households in geographically distant areas—such as Detroit, MI and Cleveland, OH—may experience correlated residuals because they are both exposed to the same global automotive industry shocks. I implement the exposure-robust inference procedure (AKM) to re-center inference on the industry-level shifts rather than the regional shares. As shown in Table A8, the primary findings for female labor withdrawal and male market intensity in dual-earner households remain statistically significant under this inference regime. For men, the results are more sensitive to this inference; while the increase in market work remains large for two-earner households, the coefficients in the broader samples are no longer statistically significant at conventional levels.

Falsification test: Controlling for wage shares and income levels. If the reallocation of female labor is driven by the automation-induced shift in relative spousal value, then directly controlling for these wage metrics should absorb the explanatory power of the robot shock. As a formal falsification test, Table A9 re-estimates the dual-earner specification while explicitly controlling for the respondent's wage share and log weekly earnings. The results indicate a gender asymmetry. For women, conditioning on the transmission mechanism drives the coefficients on robot exposure for both market work and leisure to become statistically null. This supports the interpretation that shifting relative and absolute wages are the primary structural channels driving the female labor withdrawal.

In contrast, the male coefficients for market work and leisure remains significant. This divergence suggests that as the wife withdraws from the market to preserve the traditional hierarchy, the husband acts as a secondary worker, intensifying his labor supply to offset the loss in household income and maintain his provider status. Finally, the persistence of the effect on female childcare likely reflects that domestic time reallocation is a sticky consequence of this disruption in working arrangements, rather than a margin that adjusts linearly with the wage rate itself.

3.8 Industrial Automation vs. Trade Shocks

As an additional exercise, I evaluate the robustness of the primary findings by comparing the effects of industrial automation with those of rising import competition from China. Existing literature has established that the "China shock" exerted widespread effects on U.S.

labor markets, specifically by depressing manufacturing employment and wage levels for men (Autor et al., 2013). Furthermore, similar to the robot shock, trade-induced manufacturing decline has been shown to reduce the marriage-market value of young men, resulting in lower marriage and fertility rates (Autor et al., 2019). Figure 10 illustrates that men are disproportionately concentrated in occupations with high exposure to tradable goods industries, suggesting that they bear the initial burden of both technological and trade-related disruptions.

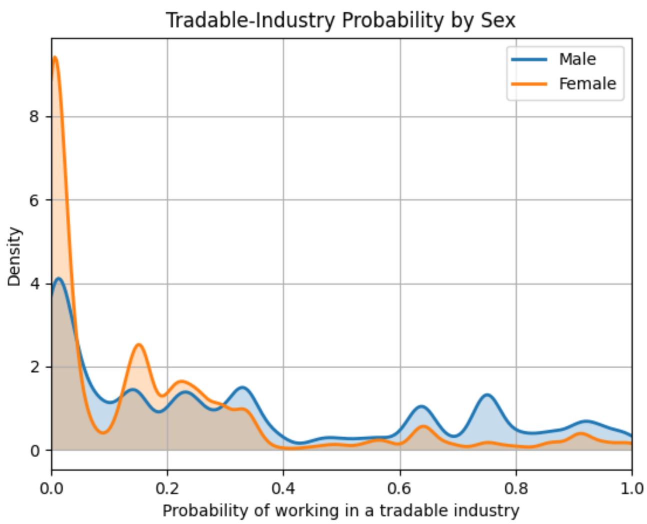


Figure 10: Kernel densities of tradable industry probability by sex. Source: United States IPUMS Census/ACS microdata, 2000–2020 (Flood et al., 2024).

Despite these shared characteristics, trade and automation operate on the household budget constraint through differing margins. Trade shocks typically manifest as industry-level contractions involving extensive-margin adjustments, such as plant closures and permanent layoffs (Halla et al., 2020). This creates a theoretical ambiguity regarding the household response. On one hand, the substantial loss of male income should activate a consumption-smoothing motive, triggering an added worker effect that compels wives to increase their labor supply to stabilize the household's financial position (Besedeš et al., 2021). On the other hand, the degradation of the husband's economic status might activate the same identity-

[3]The tradable share is computed as the pooled, person-weighted probability for each occupation o that a worker i is employed in a tradable industry:

$$\hat{p}_o = \frac{\sum_{i \in o} w_i \, \mathbb{1}\{\text{Industry}_i \in T\}}{\sum_{i \in o} w_i},$$

where T includes Agriculture/Forestry/Fishing, Mining, Manufacturing, Transport/Communications/Utilities, and Wholesale Trade.

based withdrawal observed in the context of automation, as the household attempts to preserve traditional gender roles in the face of manufacturing decline.

In contrast, industrial robots represent a task-based shock that substitutes for human labor primarily at the intensive margin, frequently allowing the firm to remain operationally viable while devaluing specific manual tasks (Acemoglu and Restrepo, 2020). Because automation depresses male potential wages and hours without necessarily resulting in the immediate total unemployment characteristic of plant closures, the insurance motive may be less prominent than the identity-maintenance motive. By comparing these two shocks, I test whether the identity mechanism generalizes to the context of trade: if trade shocks yield estimates that are directionally similar to the robot shocks, it suggests that the identity wedge is sufficiently influential to dampen or override the added worker effect even during periods of broad manufacturing contraction.

To isolate these channels and ensure the primary results are not a byproduct of concurrent globalization trends, I perform two empirical exercises. First, I replicate the baseline estimations by replacing robot exposure with commuting-zone exposure to Chinese imports, constructed following the methodology of Autor et al. (2013). This allows for a direct comparison of how the two shocks—one task-based and one trade-based—differentially affect the intra-household division of labor. Second, I perform a formal robustness check by re-estimating the primary robot-exposure coefficients while explicitly controlling for the trade shock. This joint estimation serves to verify that the impact of automation on household time use represents a structural response to a shift in the relative price of spousal time, rather than a secondary effect of the general manufacturing decline associated with international competition.

Measurement of the trade shock. The data for this empirical exercise are integrated from several sources. Detailed bilateral trade data by product and year are retrieved from the UN Comtrade Database. Domestic production and absorption values used to normalize the shocks are sourced from the U.S. Census Bureau and the NBER-CES Manufacturing Industry Database. Historical industry-specific employment at the commuting-zone level is derived from the Decennial Census/ACS microdata and County Business Patterns (CBP), following the regional definitions and industry concordances established by Autor et al. (2013).

U.S. imports from China underwent substantial expansion since the early 1990s, a trend driven by internal market reforms, large-scale rural-to-urban migration within China, and enhanced access to foreign technology. These transformations, culminating in China's accession to the WTO in 2001, which granted it permanent normal trade relations status, increased its export capacity. As shown in Figure 11, imports from China rose through the 1990s, entered a phase of rapid growth following WTO accession, and stabilized around 2014.[4]

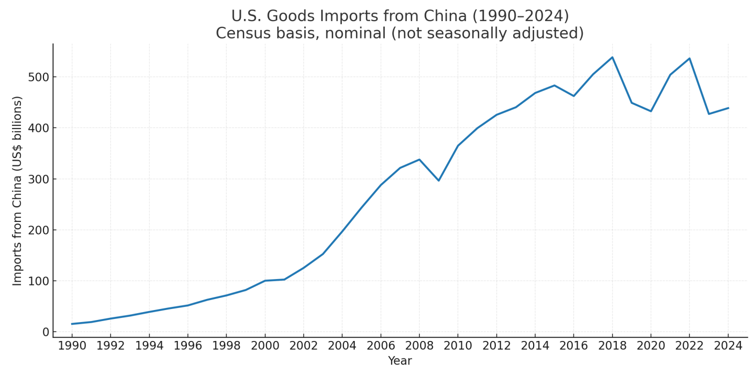


Figure 11: U.S. goods imports from China, 1990–2024 (Census basis, nominal).

Following Autor et al. (2013), I extract annual bilateral export flows from China by product to the United States and to a set of other high-income destinations. Product-level values are mapped to U.S. industries using standard NAICS concordances and aggregated to construct industry-level Chinese import exposure. To construct the measure of local exposure to import competition, I employ a shift-share methodology where the shock captures the cumulative growth in Chinese import penetration within a commuting zone, weighted by the zone's historical industrial composition in 1990. Specifically, industry-level imports from China are mapped to 397 four-digit SIC manufacturing industries. The local exposure is defined as:

$$\text{Trade}_{ct} = \sum_{s} \frac{\text{Emp}_{sc,1990}}{\text{Emp}_{c,1990}} \times \frac{\Delta \text{Import}_{st}^{Ch\rightarrow US}}{\text{Absorption}_{s,1990}}, \tag{8}$$

where $\Delta\text{Import}_{st}^{Ch\rightarrow US} = \text{Import}_{st}^{Ch\rightarrow US} - \text{Import}_{s,1991}^{Ch\rightarrow US}$ represents the growth in U.S. imports from China in industry s between the base year and year t. The denominator is the sector's 1990 domestic absorption, defined as total production plus imports minus exports. This normalization ensures that the shock represents a structural change in the trade penetration ratio rather than absolute volume. I rely on the change in exposure relative to the baseline to isolate the structural displacement affecting labor demand, a strategy analogous to the robot exposure measure defined in Equation 3.[5]

[4]Consequently, the ATUS sample used to investigate the effect of import competition is restricted to the period 2003–2014 to align with the peak years of the trade disruption.

[5]Unlike the trade measure, the robot exposure variable does not require explicit differencing from a base year because the operational stock of industrial robots in the early 1990s was negligible across most industries; thus, the contemporaneous stock is effectively equivalent to the cumulative adoption since the beginning of the period.

To address the potential endogeneity of U.S. import penetration—specifically the concern that import growth might reflect U.S. domestic demand shocks rather than Chinese supply factors—I instrument U.S. exposure using the industry-level growth of Chinese exports to eight other high-income economies: Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland. The use of these alternative markets isolates the supply-driven component of China's export growth, such as productivity gains and falling trade costs, that is orthogonal to U.S. economic conditions. The instrument Trade_{ct}^{IV} replaces the U.S. import term with these foreign import flows and utilizes 1980 employment shares as weights. The use of 1980 shares further ensures that the measure is not contaminated by anticipation effects or mechanical correlations with the labor market adjustments occurring in the 1990s and 2000s.

Gender-oriented decomposition. A central aspect of the trade shock is its variation across sectors with differing gender intensities. While industrial automation is identified by task content, trade exposure can be decomposed to isolate shocks that load onto male-dominated versus female-dominated industries. This allows for a more direct comparison with the male-biased nature of robot adoption and its effect on the intra-household division of labor. Using the historical female share of employment in each industry-commuting zone pair, $f_{cs,1990}$, and its complement, $1 - f_{cs,1990}$, I decompose the overall exposure into gender-specific shocks:

$$\text{Trade}_{ct}^{g} = \sum_{s} s_{cs,1990}^{g} \times \frac{\text{Emp}_{sc,1990}}{\text{Emp}_{c,1990}} \times \frac{\Delta\text{Import}_{st}^{Ch\to US}}{\text{Absorption}_{s,1990}}, \quad g \in \{f, m\}, \tag{9}$$

where $s_{cs,1990}^{f} = f_{cs,1990}$ and $s_{cs,1990}^{m} = 1 - f_{cs,1990}$. This decomposition identifies two distinct sources of variation: Trade_{ct}^{m} targets sectors such as heavy manufacturing and basic metals where men prevail, while Trade_{ct}^{f} loads on sectors like textiles and electronics with higher female employment shares.

Table 6 reports summary statistics for these aggregate and gender-specific trade measures. The aggregate exposure to Chinese import competition (Trade_{ct}) has a mean of 0.006 and a standard deviation of 0.010. Given the construction of the variable as a weighted average of industry-level penetration ratios, a one-standard-deviation increase corresponds to approximately a 1 percentage point rise in the share of Chinese imports in domestic absorption. Decomposing this exposure reveals the gendered nature of the disruption. The male-specific component (Trade_{ct}^{m}), which isolates exposure in male-dominated industries, averages 0.003. In contrast, the female-specific component (Trade_{ct}^{f}) is slightly smaller, with a mean of 0.002. This variation provides the necessary data to test whether trade-induced changes in the relative market value of spouses trigger the same identity-based time-use responses as those observed with industrial automation.

Table 6: Summary statistics of trade exposure measures.

Variable	Mean	Std. Dev.	Min.	Max.	N
Trade_{ct}	0.006	0.010	0	0.163	168,414
Trade_{ct}^{m}	0.003	0.005	0	0.059	168,414
Trade_{ct}^{f}	0.002	0.004	0	0.044	168,414

China shock results. Table A10 provides the initial set of comparative results using the aggregate Chinese import penetration measure. In contrast to the estimates for robot exposure, which show a diverging response between spouses, rising import competition is associated with a simultaneous reduction in market hours for both partners on the intensive margin. Specifically, the 2SLS estimates for the two-earner sample (Column 3 for men, Column 6 for women) indicate that a 1 percentage point increase in import penetration is accompanied by a reduction in male market work of approximately 51 minutes per day and a reduction in female market work of 29 minutes.

Despite this general contraction in market hours, the shift toward domestic responsibilities remains gendered: women show a significant increase in their childcare time (approximately 21 minutes per day in Column 6), whereas the effect for men is smaller and lacks statistical precision. This pattern suggests that while trade shocks function as a generalized negative wealth shock that depresses labor supply across the board—driven largely by the extensive-margin loss of manufacturing jobs—the internal household response defaults to a traditional division of labor, with women providing the bulk of compensatory childcare.

The decomposition into Trade_m and Trade_f in Table A11 allows for a more granular evaluation of the identity-norm mechanism. The male-specific trade shock (Trade_m) serves as a specular benchmark for the robot analysis, as it disproportionately targets male-dominated manufacturing hubs and thereby raises the wife's relative potential wage share. Under Trade_m, the dynamics mirror the automation results: men in two-earner households experience a significant reduction in their leisure time (between 147 and 176 minutes per 1 percentage point increase), which is consistent with an effort to maintain their absolute household contribution as their sector declines. Conversely, the results for Trade_f—which loads on sectors where women prevail—show a reversal. In this case, the shock to the wife's relative earnings is accompanied by a large and significant withdrawal of the husband from market work (approximately 277 minutes per 1 percentage point increase) and an increase in his leisure time.

Joint estimation of robot and trade shocks. As a final robustness check, I estimate a horse-race specification that includes both robot exposure and Chinese import competition simultaneously. This strategy isolates the impact of task-based automation while holding constant the broader effects of manufacturing decline and globalization. Table A12 presents these results.

For market work and leisure, the coefficients on robot exposure are consistent with the baseline estimates. In the 2SLS specification for two-earner households (Column 3), the positive effect of robots on male market work is estimated at 36.37 minutes, compared to 18.02 in the baseline. This suggests that trade and automation may exert opposing pressures on male labor supply: while trade eliminates sectors, automation devalues specific tasks while potentially requiring increased human effort in remaining roles. For women, the negative effect of robots on market work remains stable in magnitude. It retains statistical significance in the broader sample of households (Column 4, coefficient -15.69), though it shows less precision at the intensive margin (Column 6), even as the point estimate remains large at -16.63. The shift in leisure—a reduction for men and an increase for women—is preserved and remains statistically significant (Columns 3 and 6).

However, the results for childcare time are less robust in the joint estimation. While the baseline specification indicated a shift toward female childcare, the coefficients in Table A12 are smaller in magnitude and statistically insignificant for both spouses. This sensitivity indicates that the specific shift of non-market time toward childcare is difficult to disentangle from the broader disruptions associated with trade shocks, or that the high spatial correlation between the two shocks increases the standard errors. Nevertheless, the stability of the results for market work and leisure supports the finding that automation shifts intra-household labor supply toward a traditional gendered allocation independently of concurrent trade patterns.

Beyond the diverging margins of adjustment, the relative noise in the trade estimates likely reflects the conceptual distinction between aggregate regional disruption and localized productivity shifts. As documented by Autor et al. (2013) and Dauth et al. (2021), trade shocks frequently trigger general equilibrium spillovers—including local housing market declines and a collapse in non-tradable service demand—that exert a simultaneous negative wealth effect on both spouses. These ripple effects introduce aggregate noise that can obscure the identification of specific intra-household substitution patterns. In contrast, industrial robots primarily function as a scale-expanding productivity change rather than a force for sectoral elimination (Adachi et al., 2024). Consequently, automation provides a clearer identification of the shift in the intra-household relative price of time. Furthermore, the dollar-value flows used to construct trade exposure are more sensitive to exchange rate volatility and reporting lags than the physical installation of robot units, which contributes to the lower precision of the trade estimates.

The following section proposes a formalization of this causal intuition through a structural collective model of household time allocation, with parameters estimated to match the time-use patterns observed in the ATUS data.

4 A structural household model of time use

The preceding empirical analysis demonstrates that industrial automation systematically alters intra-household time allocation, but it captures only the net behavioral response. To understand the extent to which these effects are driven specifically by gender identity norms, this section develops and estimates a structural household model.

A standard frictionless model predicts that a negative shock to male earnings should trigger an added-worker effect, prompting wives to increase their market hours to smooth household consumption. The data, however, show a withdrawal. The structural framework is designed to explain why this standard income effect is overridden. It does so by embedding a breadwinner friction into the household's joint decision-making process.

Estimating this model allows for the construction of a norm-neutral counterfactual. By comparing the benchmark model against a baseline where identity constraints are absent, it can isolate the specific role of the identity penalty from standard shifts in aggregate income. Ultimately, this approach rationalizes the non-monotonic time-use profiles documented in Section 2.1 and quantifies the efficiency losses that arise when couples divert resources to preserve traditional gender roles.

4.1 Model set-up

The analysis employs a semi-cooperative household framework with the introduction of an identity friction. This specification is designed to accommodate two empirical regularities: the prevalence of corner solutions in daily childcare provision—where one parent reports zero hours—and the behavioral trend reversal at the equal-earnings threshold.

To capture the first regularity, the semi-cooperative structure treats market hours as contractible (jointly chosen) but models childcare as a privately provided public good determined non-cooperatively. Because the utility function is logarithmic and child quality production is Cobb-Douglas, the cross-partial derivative of spousal childcare time is exactly zero. Consequently, corner solutions in childcare emerge naturally whenever a spouse's shadow price of time, which is set by their first-stage market hours, strictly exceeds their private marginal benefit of care.

To address the second regularity—the non-monotonicity at earnings parity—an endogenous identity penalty enters the household's budget constraint. Highly flexible collective models can theoretically force trend reversals through complex, non-linear shifts in the Pareto weight. The present framework instead formalizes the breadwinner norm directly as a resource friction, reproducing the observed behavioral shifts with disciplined, minimal structural assumptions.

A household consists of two members, $i \in \{f, m\}$, each endowed with one unit of time, earning a market wage rate w_i, and jointly caring for n children. Spouses engage in a two-stage game. In the first stage, they cooperatively choose their optimal market labor supply; in the second stage, each individually chooses the time allocated to childcare.

Time budget. Each individual faces the time constraint:

$$h_i + l_i + (t_i + \tilde{t}_i)\, n \leq 1, \quad h_i > 0,\ l_i > 0,\ t_i \geq 0,\ \tilde{t}_i > 0, \tag{10}$$

where h_i denotes market work time, l_i leisure time, t_i parental quality time invested per child, $\tilde{t}_i$ the fixed (non-quality) childcare time cost per child, and n the number of children. Note that while market work and leisure are assumed strictly positive, quality time t_i allows for corner solutions ($t_i \geq 0$).

The quality of children is determined by a Cobb-Douglas production function:

$$q = (1 + t_f)^{\alpha}\, (1 + t_m)^{1-\alpha}, \tag{11}$$

with $\alpha \in [0, 1]$ governing the relative effectiveness of maternal versus paternal time inputs. Following Del Boca et al. (2013), consumption goods are assumed to have a negligible effect on child quality; thus, only time inputs enter the production function, a specification that facilitates the emergence of corner solutions in optimal childcare supply.

Utility and the identity wedge. Individual i's baseline utility is given by:

$$u_i \;=\; \log(c) \;+\; \mu_i \log(l_i) \;+\; \gamma_i \log(q\, n). \tag{12}$$

Following the social-friction frameworks of Greenwood et al. (2016) and Doepke and Kindermann (2019), household consumable income is adjusted by an endogenous **identity-**

maintenance friction, $\phi(s)$, which operates as a wedge on the wife's effective earnings:

$$c \ = \ \phi(s)\, w_f h_f \ + \ w_m h_m, \qquad s \ = \ \frac{w_f}{w_f + w_m}. \tag{13}$$

Although relative-status concerns frequently appear as additive costs in the utility function (Bertrand et al., 2015), the budget-wedge approach is preferable for two primary reasons.

First, $\phi(s)$ represents the endogenous resource cost of maintaining a traditional breadwinner hierarchy when the wife's potential market value approaches or surpasses the husband's. Within this framework, the threat to the traditional status distribution is driven by the relative potential wage s, and the household responds through the adjustment of market hours h_f. In line with the literature on "gender display" (Brines, 1994; Bertrand et al., 2015), the withdrawal from market work functions as a compensatory mechanism. By reducing her hours as her relative wage rises, the wife ensures that total household earnings remain aligned with traditional norms, despite her increasing market productivity. The wedge $\phi(s)$ captures the consumable utility forfeited to this status-preservation motive. It reflects the redirection of resources toward compensatory behaviors or the transaction costs of domestic friction triggered when the husband's provider status is threatened.

Second, specifying the identity wedge within the budget constraint is technically necessary to account for the failure of the standard income effect observed in the American Time Use Survey. Because $\phi(s)$ scales the hourly return to labor, it directly alters the marginal rate of substitution (MRS) between consumption and leisure. As the wife's relative potential wage w_f increases, the declining $\phi(s)$ functions as an endogenous marginal tax. This identity penalty depresses the net marginal return to female labor so severely that it dominates the standard consumption-smoothing income effect. It forces the wife to reduce her market hours even though the household is absolutely poorer. This mechanism replicates the sharp non-monotonicity and subsequent withdrawal from the labor market documented in Section 2.1, a pattern that a separable utility cost struggles to generate. Furthermore, this formulation distinguishes the identity friction from standard shifts in intra-household bargaining power. While endogenizing the Pareto weight as an increasing function of the wife's wage share, $\theta(s)$, successfully generates the monotonic rise in female leisure observed empirically, continuous shifts in bargaining power cannot endogenously produce the localized, sharp trend reversals in market work and childcare precisely at the parity point ($s = 0.5$). By holding the structural bargaining weight fixed across the distribution and introducing the identity wedge, the model explicitly isolates norm maintenance as the driver of the non-monotonicity.

Accordingly, the identity-maintenance friction is specified as a nonlinear function of the wife's relative potential wage share s:

$$\phi(s) \;=\; \exp\!\big[-\chi\, s^{\kappa}\big], \quad \chi > 0,\ \kappa > 1.$$

The curvature parameters χ and κ ensure that social costs materialize primarily as the wife's market value crosses a threshold, while keeping consumption strictly positive. This continuous formulation avoids imposing a discrete penalty at an arbitrary cutoff. Relying on a smooth, flexible function allows the data to endogenously determine the onset and intensity of the friction. Additionally, a continuous and differentiable $\phi(s)$ maintains the tractability of the household's optimization problem, guaranteeing a well-defined marginal rate of substitution and a unique equilibrium in the choice of market hours.

Stages. Agents play a two-stage semi-cooperative game à la Cournot, solved by backward induction.

Stage 2 (Childcare): Taking market hours h_i as given from the first stage, each individual i maximizes their private utility with respect to t_i:

$$v_i(t_i, t_{-i} \mid h) = \log(c) \;+\; \mu_i \log\!\big(1 - h_i - (t_i + \tilde{t}_i)\, n\big) \;+\; \gamma_i \log\!\big((1+t_f)^{\alpha}(1+t_m)^{1-\alpha} n\big). \quad (14)$$

Because the child quality production function is Cobb-Douglas and utility is logarithmic, the objective function is additively separable in t_f and t_m.[6] This stage results in four possible regimes: both, one, or neither spouse provides quality childcare time.

Stage 1 (Market Work): In the first stage, agents cooperatively choose market hours to maximize the household utility function, anticipating the Nash equilibrium outcomes of the second stage. The household objective is the weighted sum of individual utilities:

$$u_h \;=\; \theta\, u_f \;+\; (1-\theta)\, u_m,$$

where $\theta \in (0, 1)$ denotes the bargaining power of the wife.

Under mild regularity conditions, this game admits a unique subgame perfect equilibrium. Since each spouse's time-allocation choice in Stage 2 solves a strictly concave maximization

[6]Specifically, the cross-partial derivative of spousal childcare time is zero ($\frac{\partial^2 v_i}{\partial t_i \partial t_{-i}} = 0$), implying orthogonal reaction functions. While this eliminates strategic substitutability, a structural "lack of commitment" friction remains (Gobbi, 2018). In the semi-cooperative case, the FOC for spouse f is $\frac{\alpha\gamma_f}{1+t_f} = \frac{\mu_f n}{l_f}$. In a fully cooperative (Pareto-efficient) benchmark maximizing $\theta v_f + (1-\theta) v_m$, the FOC becomes $\frac{\alpha\gamma_f}{1+t_f} + \frac{(1-\theta)}{\theta}\frac{\alpha\gamma_m}{1+t_f} = \frac{\mu_f n}{l_f}$. Because the social marginal benefit in the cooperative case includes the internalized positive externality to the partner (γ_m), the semi-cooperative Nash equilibrium leads to a structural under-provision of childcare ($t_i^{semi} < t_i^{coop}$).

problem over a compact, convex set:

$$\{t_i \geq 0: \; h_i + l_i + (t_i + \tilde{t}_i)n \leq 1\},$$

Rosen's existence theorem ensures a unique Nash equilibrium in childcare for any given pair of market hours. In Stage 1, the induced household payoff is maximized. Since the utility function is continuous in (h_f, h_m) and diagonally strictly concave, Brouwer's fixed-point theorem ensures existence, while the diagonal strict concavity ensures the uniqueness of the profile $(h_f^*, h_m^*, t_f^*, t_m^*)$. Thus, the model admits exactly one equilibrium. Depending on the second-stage outcome, the solution falls into one of the twelve profiles reported in Appendix Section C.

Assumptions. A1 (Primitives). Preferences are defined as in (12) with parameters $\mu_i > 0$, $\gamma_i > 0$, $\alpha \in (0,1)$, wages $w_i > 0$, number of children $n \in \mathbb{N}$, and fixed time costs $\tilde{t}_i \geq 0$ such that the time budget is feasible $(1 - n\tilde{t}_i > 0)$ for $i \in \{f, m\}$.

A2 (Norm wedge). The function $\phi : [0,1] \to (0,1]$ is C^2, strictly decreasing, and log-convex in s (e.g., $\phi(s) = \exp[-\chi s^\kappa]$ with $\chi > 0, \kappa > 1$). Because s is defined on exogenous potential wage shares $(s = w_f/(w_f + w_m))$, $\phi(s)$ acts as a constant parameter during the household's time-allocation optimization. Consequently, household consumable income c is strictly linear in (h_f, h_m), ensuring the mapping $(h_f, h_m) \mapsto \log(c)$ is strictly concave on the feasible set.

A3 (Feasible sets). For any (h_f, h_m) such that $0 \leq h_i \leq 1 - n\tilde{t}_i$, each player's second-stage choice set for t_i is nonempty, convex, and compact. The joint feasible set for (h_f, h_m, t_f, t_m) is also nonempty, convex, and compact.

Proposition 1 (Existence and uniqueness)**.** *Under Assumptions A1–A3:*

(i) ***Stage 2 (Childcare).*** *For any pair of market hours (h_f, h_m), the second-stage game in (t_f, t_m) is a diagonally strictly concave game. Therefore, it admits a* unique *Nash equilibrium (t_f^*, t_m^*), and the equilibrium correspondence $(t_f^*, t_m^*)(h_f, h_m)$ is single-valued and continuous.*

(ii) ***Stage 1 (Market work).*** *Let $V(h_f, h_m) = \theta\, v_f\big(h_f, h_m, t^*\big) + (1-\theta)\, v_m\big(h_f, h_m, t^*\big)$ denote the household objective induced by the unique Stage 2 equilibrium. Under A2, V is strictly concave on the feasible set of (h_f, h_m); therefore, the Stage 1 problem has a* unique *maximizer (h_f^*, h_m^*).*

Consequently, the two-stage semi-cooperative problem has a unique *subgame-perfect equilibrium $(h_f^*, h_m^*, t_f^*, t_m^*)$.*

4.2 Estimation

The model is estimated to replicate observed patterns in women's market work, childcare time, and their share of household wages. Wages and the number of children are drawn from log-normal distributions parameterized using summary statistics from the ATUS sample (Table 7).[7]

The distributions for female wages, male wages, and the number of children are drawn independently. Modeling these variables as orthogonal serves a specific identification purpose: it ensures that the non-monotonic time-use profiles generated by the simulation are driven strictly by the relative wage share activating the identity friction, rather than by mechanical demographic correlations—such as high-wage women mechanically supplying less childcare due to having fewer children. Furthermore, properly accounting for the endogenous joint determination of fertility and human capital would require a dynamic life-cycle framework, which falls outside the scope of this static time-allocation model.

Table 7: Summary statistics for simulation inputs

Variable	Mean	Std. Dev.
Female wages (w_f)	16.309	9.366
Male wages (w_m)	18.904	10.084
Number of children (n)	2.105	0.978

The bargaining power parameter, θ_i, is assumed to follow a logistic-normal distribution:

$$\theta_i \;=\; \frac{1}{1+\exp(-z_i)}, \qquad z_i \;\sim\; \mathcal{N}\big(\bar{\theta},\; \sigma^2\big),$$

which ensures $\theta_i \in (0, 1)$. Here, $\bar{\theta}$ and σ^2 denote the mean and variance of the underlying normal distribution, respectively.

The preference weight for consumption is normalized to unity. Under logarithmic utility, only the ratios between the preference weights on consumption, leisure, and child quality are identified by the time-use moments; scaling all utility weights by a constant leaves choices unchanged. Because consumption quantities are unobserved in the data, fixing the consumption weight anchors the utility scale and prevents weak identification. The child-quality productivity parameter, α, is fixed at 0.5, following Del Boca et al. (2013) and Gobbi

[7]Nominal wages are deflated to constant-2000 dollars.

(2018). The remaining parameters,

$$\Theta = (\gamma_f,\, \gamma_m,\, \mu_f,\, \mu_m,\, \tilde{t}_f,\, \tilde{t}_m,\, \bar{\theta},\, \sigma,\, \chi,\, \kappa),$$

are estimated via the Method of Simulated Moments (MSM). Formally, the estimator minimizes the weighted distance between data and model moments:

$$\hat{\Theta} = \arg\min_{\Theta} \left(\frac{m - \hat{m}(\Theta)}{m}\right)' W \left(\frac{m - \hat{m}(\Theta)}{m}\right), \tag{15}$$

where m stacks the mean shares of time allocated to market work and childcare across five bins of the wife's relative wage share, $s = \frac{w_f}{w_f + w_m} \in (0.25, 0.75)$, yielding 20 moments in total. Note that while the empirical analysis documents the response of leisure, it is explicitly excluded from the set of targeted moments. Because the time budget constraint strictly binds ($l_i = 1 - h_i - (t_i + \tilde{t}_i)n$), leisure is structurally determined as the residual time category. Simultaneously targeting market work, childcare, and leisure would introduce perfect multicollinearity into the moment conditions; hence, the model is fully identified by targeting only the linearly independent non-leisure time uses.

I employ the identity weighting matrix, $W = I$, for three primary reasons: (i) the targeted moments are standardized as percentage deviations, making equal weighting effectively close to efficient; (ii) ATUS diary measures are zero-inflated and heteroskedastic, with sparsity in some bins making the empirical covariance matrix noisy and potentially ill-conditioned; and (iii) the identity matrix yields a smoother objective function, improving the stability of the optimization routine. The minimization is performed in two stages: a Differential Evolution search provides a global candidate minimum, which then seeds a local quadratic-approximation routine.

Table 8 presents the parameter estimates obtained by simulating 50,000 households, both for the full model ($\phi \neq 1$) and the restricted model without the social norm ($\phi = 1$). In the baseline specification, the logistic-normal parameters ($\bar{\theta} = -0.45$) imply an average female Pareto weight of approximately 0.43. This estimate is consistent with the broader collective household literature, which typically estimates female bargaining weights below 0.5 (Knowles, 2013; Friedberg and Webb, 2006).

In contrast, estimating the model without the identity wedge ($\phi = 1$) yields a positive mean for the underlying distribution ($\bar{\theta} = 0.35$), implying an average female Pareto weight of approximately 0.56. This counter-intuitive result indicates that, in the absence of the identity mechanism, the model must attribute an unrealistically high bargaining power to women to rationalize their observed labor supply choices. By explicitly modeling the identity cost, the benchmark model recovers bargaining parameters that align more closely with external empirical evidence, providing further support for the identity mechanism as a driver of intra-household allocation.

Table 8: Structural parameter estimates

Parameter	Description	Estimates	
		Benchmark ($\phi \neq 1$)	No Norm ($\phi = 1$)
γ_f, γ_m	Preferences for child quality	3.67, 4.48	1.49, 4.02
$\tilde{t}_f, \tilde{t}_m$	Fixed childcare time cost	0.09, 0.05	0.08, 0.06
μ_f, μ_m	Preferences for leisure	1.19, 1.11	1.15, 1.37
$\bar{\theta}, \sigma$	Bargaining power (z_i distrib.)	-0.45, 1.98	0.35, 1.75
κ, χ	identity wedge parameters	4.19, 7.27	–

4.3 Numerical results

This section evaluates the model's fit by comparing the simulated moments to the empirical targets, followed by counterfactual simulations that illustrate household responses to wage shocks across the income distribution.

4.3.1 Comparison with data

Table 9 contrasts the empirical averages for market work, childcare time, and employment rates against the simulated moments generated under the benchmark parameterization (Table 8). The simulation closely tracks the observed time-allocation distributions. Men devote roughly half of their available time budget to market labor and 12% to childcare, aligning almost precisely with the empirical targets. For women, the framework accurately captures the domestic division of labor, generating a market work share of 26% (compared to the observed 24%) and a childcare share of 22% (against 23% in the data).

Evaluating the identity-neutral specification ($\phi = 1$) demonstrates the structural necessity of the breadwinner friction. Absent the identity norm, the estimation fails to reconcile the gendered division of labor with baseline participation rates. The restricted model systematically underpredicts male market work (0.44 versus 0.47) and overpredicts female market work (0.30 versus 0.24). In addition, the benchmark simulation yields a high male employment rate (99.9%) and a distinctly lower female employment rate (57%). This outcome qualitatively reproduces the empirical employment gap (97% versus 68%), which is notable given that participation rates serve as untargeted external moments in the estimation.

Table 9: Comparison of model and data averages

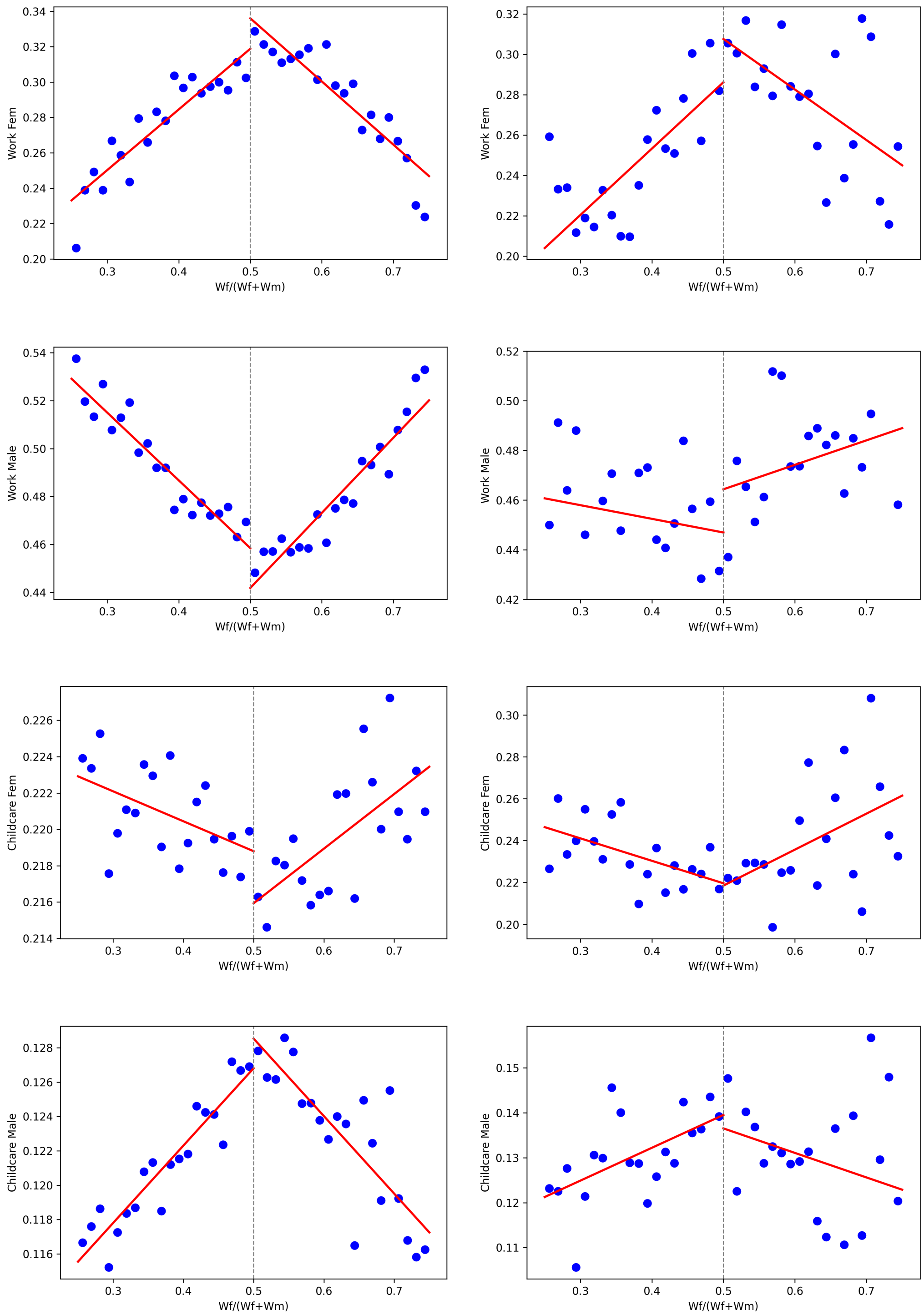
Sex	Moment	Model mean ($\phi \neq 1$)	Model mean ($\phi = 1$)	Data mean
Internal moments (Targeted)				
Male	Work	0.4989	0.4360	0.4709
	Childcare	0.1207	0.1303	0.1280
Female	Work	0.2608	0.3006	0.2443
	Childcare	0.2210	0.1755	0.2342
External moments (Untargeted)				
Male	Employment	0.9999	0.8543	0.9715
Female	Employment	0.5663	0.7163	0.6775

Figure 12 maps the simulated time-use profiles for market hours and childcare across five intervals of the wife's wage share, plotted alongside the corresponding ATUS data. The structural framework endogenously reproduces the primary non-monotonicities documented in Section 2.1. In particular, it generates the inverse-V trajectory in female labor supply: market hours expand as the wife's relative wage converges toward parity, yet they strictly contract once she crosses the equal-earnings threshold. By the same mechanism, the simulation mirrors the V-shaped trough in maternal childcare time and the concurrent upward spike in male market effort that materializes once the wife becomes the primary earner. This alignment between the simulated and empirical profiles verifies that the estimated identity penalty operates as a sufficient friction to drive the kinked behavioral adjustments isolated in the reduced-form analysis.

4.3.2 Counterfactuals

Empirical studies of the U.S. robot wave show that the observed narrowing of the gender wage gap is driven primarily by declining earnings—or outright job loss—among men in the bottom-to-middle terciles. Conversely, wage compression on the female side arises from gains concentrated in the upper terciles of the female distribution (Acemoglu and Restrepo, 2020; Cortes et al., 2024).

To replicate this pattern quantitatively, I consider wage shocks targeting the bottom tercile of the male wage distribution. A challenge in calibrating these shocks is that empirical estimates typically measure effects on observed wages among employed workers. However, in the structural model, agents respond to changes in their potential market wage, which governs decisions on both the intensive (hours worked) and extensive (participation) margins.



Model **Data**

Figure 12: Simulated vs. observed time shares for work and childcare across the distribution of the wife's wage share.

Automation Shock. For the robot counterfactual, I simulate a "high-exposure" scenario of 3 additional robots per 1,000 workers. This magnitude represents the "Rust Belt" experience (e.g., commuting zones in Michigan and Ohio), which drove the aggregate displacement results in Acemoglu and Restrepo (2020).

Using the 3-year estimate of $\beta_{wage} = -0.061$, this exposure corresponds to an 18.3% decline (3×-0.061) in potential wages for men in the first tercile, or a male wage multiplier of approximately 0.82. I evaluate the household's response to these shocks across the range of multipliers shown in Figure 13.

Female Wage Complementarity. In addition to the negative shocks to male earnings, it is reasonable to expect an accompanying positive variation in the potential wages of women in the upper tail of the distribution. Unlike the direct displacement experienced by low-skill men, the effect on high-skill women operates through occupational upgrading and sectoral reallocation.

Cortes et al. (2024) find that in automation-exposed labor markets, women are significantly more likely than men to transition out of routine tasks and into cognitive, high-wage occupations. While the counterfactuals presented below conservatively isolate the decline in male opportunity costs, these studies suggest a complementarity premium for women—potentially in the range of 2%—which would act as a mirroring force, accelerating the convergence of spousal relative earnings and amplifying the identity mechanism.

Simulation Results. To isolate the role of the identity penalty, I compare the responses across two model regimes: the identity-neutral baseline ($\phi = 1$) and the benchmark model.

In the identity-neutral regime, the model isolates the aggregate income effect: a 20% negative shock to male wages (multiplier 0.8) triggers an added-worker effect. Because the household is poorer, women increase their market work to smooth consumption, causing the gender hours gap ($h_m - h_f$) to contract by approximately 11.1% (falling from 0.135 to 0.120).

However, in the benchmark model, this insurance mechanism fails. Because the exogenous shock depresses the husband's potential wage, the wife's relative potential wage share (s) rises. This pushes households near the threshold into the region where the identity penalty binds. The wedge induces a structural withdrawal of women from the market to maintain traditional status. For the same 20% shock, the gender work gap in the norm case increases by 4.2% (rising from 0.238 to 0.248). This behavioral reversal means that at the 0.8 multiplier, the gender hours gap is 107% larger than it would be in an identity-neutral baseline.

Figure 13 illustrates these dynamics for both work and childcare. The right panel displays the response of the gender gap in childcare ($t_f - t_m$). In the identity-neutral benchmark, the gap contracts by 1.5% as the husband's lower opportunity cost of time encourages his participation in domestic tasks. In contrast, under the identity norm, the childcare gap increases by 0.4% (rising from 0.1004 to 0.1008). While the husband's market value falls, the wife's compensatory specialization in home production intensifies to preserve the provider hierarchy. Consequently, the childcare gap in the benchmark model is 124% higher than in the neutral case (0.1008 vs. 0.0450).

The "Difference" (purple dashed line) highlights the escalating cost of the norm. As automation pushes couples toward parity, the identity distortion in market hours nearly doubles, indicating that households willingly forgo consumption smoothing to maintain traditional breadwinner identities.

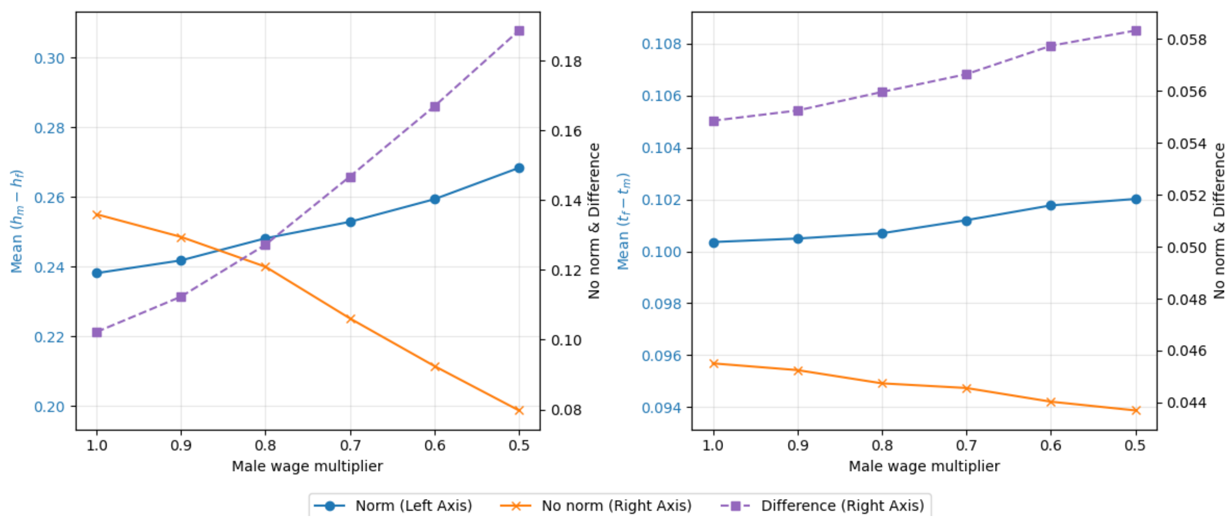


Figure 13: Simulated divergence in time use gaps in response to negative shocks to first-tercile male wages. Values for the Norm case (blue line) are on the left vertical axis. Values for the No Norm case (orange line) and the Difference (purple line) are on the right vertical axis. The x-axis represents the multiplier applied to male wages.

The structural simulation of the identity-norm regime produces a behavioral pattern that is consistent, both in sign and in the relative sensitivity of adjustment margins, with the causal estimates reported in Table 3. To facilitate a direct comparison between the absolute minute changes in the 2SLS estimates and the structural model's share-based results, I normalize the point estimates by the sample means from Table 1. In the empirical analysis, a 3-unit robot shock (corresponding to the 18.3% potential wage decline) implies a structural reduction in female market hours of approximately 45.8 minutes per day, which represents a 34.6% reduction relative to the female sample mean of 132.2 minutes. While the structural model provides a conservative lower bound—predicting a 6.2% withdrawal in time shares—it replicates the directional failure of the added-worker effect and the magnitude of the identity-maintenance friction.

Consistent with the empirical findings, the model also captures the relatively weaker response of childcare time compared to market labor. Normalizing the 2SLS results for childcare (8.5 minutes per robot in column 6) suggests that a 3-unit shock increases female childcare by 17.7% relative to the sample mean of 142.9 minutes. The ratio between the market work contraction and the childcare expansion in the data is approximately 2:1, a hierarchy of adjustment margins that is mirrored by the structural model. This alignment between the 2SLS estimates and the simulated norm regime reinforces the interpretation of industrial automation as a primary driver of the non-monotonic time-use profiles observed in the American Time Use Survey.

5 Conclusion

This paper examines how industrial automation reshapes the allocation of time within households. By linking ATUS time diaries to commuting-zone exposure to robots and addressing endogeneity with external industry shifters, I document systematic within-couple reallocations. Greater exposure to robots induces women to shift time out of market work and into childcare and leisure, while men simultaneously increase market work and reduce leisure. Quantitatively, one additional robot per 1,000 workers reallocates roughly one hour per week at the household level across work, childcare, and leisure.

These empirical patterns are rationalized by a structural household model featuring an identity wedge that penalizes couples when the wife out-earns the husband. The model reproduces the observed non-monotonicity in time use: as the wife's potential wage share rises toward parity, her market hours increase; however, once the breadwinner threshold is crossed, the identity cost triggers a withdrawal from the labor market and an increase in childcare time. Counterfactual simulations indicate that this norm roughly doubles the gender gap in market hours relative to an identity-neutral benchmark. Because automation shocks disproportionately depress men's wages in the lower-middle of the distribution, they push a significant mass of couples into the region where this norm binds, amplifying gender gaps in time use even as the raw wage gap narrows.

The findings highlight that cultural norms act as a substantial friction, preventing the efficient allocation of time following economic shocks. This suggests specific avenues for policy intervention.

First, the results provide an argument for targeted wage insurance. Standard adjustment assistance often focuses on retraining displaced workers. However, this paper shows that a drop in the husband's wage can trigger a secondary withdrawal of the wife from the labor force. Wage insurance schemes that stabilize the earnings of displaced male workers could prevent the decline of female labor supply by keeping households away from the parity threshold where the identity penalty binds.

Second, place-based policies must account for the gendered nature of local demand shocks. Regional development policies often aim to restore manufacturing jobs. Given the structural shift toward services—where women frequently hold a comparative advantage—development strategies should also facilitate the transition of men into service-sector roles. Reducing the strong gender segregation of occupations would align labor demand with household supply and, over the long term, potentially soften the rigidity of these identity penalties.

In sum, automation does not only reallocate tasks across firms; it reallocates time within households. When relative wages move couples toward earnings parity, intra-household frictions can reverse progress in gender equality. Recognizing and quantifying this interaction is essential for designing policies that ensure the benefits of technological change are distributed efficiently within the family.

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Data availability The following data sources have been used for the analysis in this article:

- **American Time Use Survey (ATUS)** – U.S. Bureau of Labor Statistics, 2003–2023.
- **Current Population Survey (CPS)** – ASEC and basic monthly files, via IPUMS-CPS (Flood et al. 2024 release); 1990–2023.
- **1990 U.S. Census Microdata** – Used for baseline potential hourly wages, via IPUMS-USA.
- **Churches and Church Membership in the United States, 1990** – Association of Statisticians of American Religious Bodies (ASARB).
- **ISSP "Family and Changing Gender Roles IV"** – 2012 U.S. subsample.
- Patent-based automation-risk scores – Webb (2020, *American Economic Review: Insights* 2(3): 300-17).
- O*NET manual-task intensity scores – Autor & Dorn (2013, *AER* 103(5): 1553-97).
- 1990 commuting-zone × industry employment shares – Acemoglu & Restrepo (2020, *JPE* 128(6): 2188-2244).
- 1970 commuting-zone × industry employment shares – Decennial IPUMS data.
- Industrial-robot stocks (country × industry, 1993-2019) – International Federation of Robotics, *World Robotics* database.
- 1990 industry employment for EU-5 (used in the robot instrument) – EU KLEMS Growth & Productivity Accounts, 2017 edition.
- U.S. import-exposure, domestic absorption and baseline instrument files – Autor, Dorn & Hanson (2013, "The China Syndrome," *AER* 103(6): 2121-68).
- Chinese exports to eight alternative OECD markets – UN Comtrade extraction following Autor–Dorn–Hanson.
- Commuting-zone boundaries and county cross-walks – Autor, Dorn & Hanson replication materials (2013).

Conflict of interest The author declares that he has no conflict of interest.

Appendix

A Supplementary tables

Table A1: Descriptive statistics: ATUS general population

	Females			Males		
Variable	Obs	Mean	Std. Dev.	Obs	Mean	Std. Dev.
Leisure (min/day)	122,710	310.07	208.33	96,658	356.87	235.57
Childcare (min/day)	122,710	43.10	101.62	96,658	22.34	69.34
Market work (min/day)	122,710	126.61	217.06	96,658	191.39	261.12
Age	122,710	48.37	18.15	96,658	46.77	17.44
Hourly wage (USD)	35,769	15.08	9.23	29,080	17.52	10.25
Weekly earnings (USD)	61,698	709.03	513.95	54,518	951.10	596.42
Usual work hours per week	113,670	21.20	20.66	88,373	30.94	22.49
Less than secondary (%)	122,710	14.37	35.08	96,658	15.69	36.37
Secondary (%)	122,710	25.91	43.81	96,658	25.52	43.60
Some college (%)	122,710	27.95	44.87	96,658	25.39	43.52
Tertiary (%)	122,710	31.77	46.56	96,658	33.40	47.17

Notes: "General population" includes all ATUS respondents by sex. Variable definitions are as in Table 1.

Table A2: Effect of robots on time spent working, childcaring, and leisuring per day (2SLS). Sample: respondents with no spouse or unmarried partner.

	Males		Females	
	Work	Leisure	Work	Leisure
Robots	2.073	3.891	-0.771	4.697
	(7.765)	(4.350)	(7.150)	(5.124)
Observations	4,071	4,071	4,630	4,630
R-squared	0.063	0.053	0.035	0.029
R-squared (OLS)	0.378	0.274	0.332	0.248

Notes: Controls include commuting zone and state-by-year FE, respondent's age/education, day, month, holiday. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Effect of Robot exposure on wages and wage gap (3, 5, and 7-year differences, 2SLS).

	Male Wages	**Female Wages**	**Wage Gap**
	(1)	(2)	(3)
Panel A: 3-Year Differences			
Robots	-0.061***	-0.009	-0.050***
	(0.011)	(0.010)	(0.013)
Observations	1,223	1,257	1,141
R^2	0.040	0.012	0.027
Panel B: 5-Year Differences			
Robots	-0.025**	0.005	-0.027
	(0.010)	(0.012)	(0.018)
Observations	1,003	1,020	927
R^2	0.032	0.015	0.014
Panel C: 7-Year Differences			
Robots	-0.014**	0.004	-0.017*
	(0.006)	(0.008)	(0.009)
Observations	806	825	754
R^2	0.054	0.011	0.027

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Notes: Sample: Working age individuals (25–54) in the ATUS. The dependent variable is the long difference (3, 5, or 7 years) of the log hourly wage for men, women, or the gender gap. All specifications include 1990 baseline controls and year fixed effects. Robust standard errors clustered at the commuting-zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Robustness to Sample Selection: Effect of robots on market work, childcare, and leisure using Inverse Probability Weighting (2SLS).

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Robots	12.45**	4.831	12.90**	-14.73***	-18.38***	-17.04***
	(5.659)	(4.492)	(6.331)	(5.196)	(5.469)	(6.213)
Daily minutes childcare						
Robots	-0.589	2.086	0.377	5.898**	9.188***	8.682***
	(2.141)	(1.980)	(2.020)	(2.793)	(3.178)	(3.339)
Daily minutes leisure						
Robots	-3.784	-1.575	-8.405**	8.525**	13.55***	12.40***
	(4.512)	(3.474)	(3.509)	(3.481)	(3.451)	(4.149)
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. Control variables include commuting zone and state-by-year FE, household demographics, day/month/holiday. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Effect of robot shock on hours worked per week (2SLS).

	Males			Females		
Weekly hours worked	(1)	(2)	(3)	(4)	(5)	(6)
Robots	-0.168	-0.0430	-0.159	-1.010**	-1.345***	-1.026**
	(0.316)	(0.203)	(0.286)	(0.464)	(0.390)	(0.421)
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. Control variables include commuting zone and state-by-year FE, household demographics, day/month/holiday. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Rotemberg Weights by Industry

Industry	**Weight**	**Industry**	**Weight**
Services	0.000	Furniture	0.010
Metal Machinery	0.008	Vehicles Other	0.013
Agriculture	0.008	Mineral	0.017
Manufacturing, other	0.008	Metal Products	0.019
Metal Products	0.008	Machinery	0.023
Construction	0.009	Food	0.039
Agriculture	0.009	Petrochemicals	0.048
Mining	0.009	Metal Basic	0.081
Research	0.009	Electronics	0.084
Textiles	0.009	Manufact Other	0.122
Utilities	0.009	Automotive	0.448
Paper	0.009		

Table A7: Effect of robots on time use, adjusting for commuting-zone-specific trends across quartiles of employment share in the automotive sector (2SLS).

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Robots	14.44**	10.29**	16.99**	-11.32**	-13.49**	-10.65
	(6.968)	(4.514)	(7.935)	(5.224)	(6.300)	(6.768)
Daily minutes childcare						
Robots	1.913	2.307	1.212	3.918	9.218***	9.417***
	(2.503)	(1.976)	(2.645)	(3.094)	(2.873)	(3.644)
Daily minutes leisure						
Robots	-9.457***	-5.544*	-12.87***	7.294**	9.601***	9.024**
	(3.413)	(3.277)	(3.969)	(3.565)	(3.524)	(4.149)
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. Control variables include commuting zone and state-by-year FE, household demographics, day/month/holiday. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Effect of robots on time use (2SLS). Adão et al. (2019) test for standard errors.

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Robots	13.92	8.38	18.02***	-15.40***	-18.91***	-15.26***
	(18.82)	(22.65)	(3.21)	(5.44)	(1.87)	(4.49)
Daily minutes childcare						
Robots	0.16	1.69	-0.77	6.61	8.22	8.45***
	(1.30)	(15.66)	(6.88)	(5.27)	(6.12)	(1.43)
Daily minutes leisure						
Robots	-8.79	-3.87	-11.87	6.75***	10.13***	9.41***
	(17.21)	(9.70)	(7.89)	(1.89)	(0.48)	(0.99)
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. Control variables include commuting zone and state-by-year FE, household demographics, day/month/holiday. Standard errors calculated using the Adão et al. (2019) procedure. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Falsification test: Isolating the identity channel by controlling for relative and absolute wages.

	Daily Minutes Work		**Daily Minutes Childcare**		**Daily Minutes Leisure**	
Variable	Males	Females	Males	Females	Males	Females
Robots	19.90**	-9.559	-0.226	10.73***	-12.69**	5.777
	(7.772)	(6.866)	(3.403)	(3.045)	(5.168)	(4.039)
Respondent's wage share	-220.2***	-155.5***	34.31	72.44**	105.6***	44.25
	(49.17)	(43.62)	(22.39)	(29.80)	(34.43)	(31.22)
$\log WeeklyEarnings$	65.74***	76.65***	-13.98**	-20.93***	-26.97***	-18.04***
	(10.02)	(9.852)	(10.05)	(6.201)	(6.667)	(6.706)

Notes: Sample is restricted to dual-earner households, capturing the intensive margin of labor supply. The table formally tests the transmission channel by controlling for the respondent's wage share (s_{pot}) and absolute log weekly earnings. The complete absorption of the robot exposure coefficients for female work and leisure validates the relative-wage identity mechanism. Standard errors clustered at the commuting-zone–year level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Effect of China shock on time spent working, childcaring, and leisuring per day (2SLS).

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Trade	-6,464***	-5,944**	-5,106***	1,907	-2,062	-2,940**
	(1,809)	(2,391)	(1,550)	(1,192)	(1,380)	(1,433)
Observations	3,507	5,726	3,305	6,324	4,300	3,899
R-squared	0.431	0.431	0.512	0.272	0.402	0.460
Daily minutes childcare						
Trade	1,127	2,054	605.2	138.9	967.3	2,067**
	(873.1)	(1,477)	(825.9)	(656.8)	(795.3)	(853.1)
Observations	3,507	5,726	3,305	6,324	4,300	3,899
R-squared	0.225	0.171	0.316	0.227	0.226	0.283
Daily minutes leisure						
Trade	-389.9	469.2	-168.8	-1,453*	-784.9	-362.8
	(1,145)	(1,200)	(1,235)	(869.9)	(977.6)	(1,112)
Observations	3,507	5,726	3,305	6,324	4,300	3,899
R-squared	0.312	0.270	0.378	0.208	0.289	0.338
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample: married or cohabiting couples with at least one spouse working and children aged 10 or less. Controls include state-by-year FE, demographic/household controls, day/month/holiday FE. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Effect of decomposed China shock on time spent working, childcaring, and leisuring per day (2SLS).

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Trade_{Male}	12,104	3,581	8,834	3,027	-1,340	-4,278
	(8,036)	(5,805)	(7,418)	(5,362)	(7,632)	(8,164)
Trade_{Fem}	-34,282***	-20,546**	-27,758**	-476.7	-3,874	-1,475
	(12,443)	(9,692)	(12,004)	(8,515)	(12,102)	(13,134)
Daily minutes childcare						
Trade_{Male}	6,228	4,300	7,127*	-2,684	-1,603	-2,293
	(3,924)	(2,843)	(3,655)	(3,261)	(3,590)	(4,392)
Trade_{Fem}	-6,492	-1,309	-9,280	4,910	5,079	8,939
	(5,678)	(4,613)	(5,695)	(5,229)	(5,832)	(7,029)
Daily minutes leisure						
Trade_{Male}	-17,611***	-5,883	-14,711***	-809.3	2,043	426.1
	(5,484)	(3,835)	(5,416)	(4,216)	(4,775)	(5,188)
Trade_{Fem}	25,302***	10,115	21,618**	-2,451	-5,012	-1,888
	(8,742)	(6,177)	(8,664)	(6,495)	(7,639)	(8,234)
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. All specifications use 2SLS. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Joint effects of robots and import competition on time spent working, childcaring, and leisuring per day (2SLS).

	Males			Females		
	(1)	(2)	(3)	(4)	(5)	(6)
Daily minutes work						
Robots	30.02***	18.57**	36.37***	-15.69*	-10.44	-16.63
	(8.493)	(8.177)	(8.249)	(8.660)	(12.94)	(14.47)
Trade	-7,383***	-7,244*	-6,401***	3,887*	-16.90	-920.8
	(2,661)	(3,742)	(2,147)	(2,169)	(2,729)	(2,862)
Observations	6,795	6,582	3,905	7,668	4,974	4,546
R-squared	0.426	0.455	0.529	0.297	0.430	0.476
Daily minutes childcare						
Robots	-0.547	1.337	-0.569	-2.966	2.308	1.579
	(3.474)	(3.902)	(3.376)	(5.544)	(6.428)	(7.178)
Trade	-1,089	-295.3	-1,556	-981.2	-622.6	123.0
	(1,321)	(1,829)	(1,254)	(1,150)	(1,487)	(1,545)
Observations	6,795	6,582	3,905	7,668	4,974	4,546
R-squared	0.170	0.179	0.323	0.230	0.232	0.290
Daily minutes leisure						
Robots	-7.832	-2.453	-12.69**	8.970	12.90*	17.83**
	(7.182)	(5.219)	(5.824)	(6.480)	(7.149)	(7.701)
Trade	-110.7	558.8	7.496	-1,701	974.0	857.0
	(1,926)	(1,797)	(1,624)	(1,555)	(1,776)	(1,829)
Spouse employed	✓		✓	✓		✓
Respondent employed		✓	✓		✓	✓

Notes: Sample includes married or cohabiting couples with at least one spouse working and children aged 10 or less. Columns 1 and 4 capture the extensive margin of the respondent (spouse is employed). Columns 2 and 5 capture the extensive margin of the spouse (respondent is employed). Columns 3 and 6 restrict to dual-earner households, capturing only the intensive margin of labor supply. Control variables in all specifications include state-by-year fixed effects, demographic/household controls, day/month/holiday FE. Standard errors clustered at the commuting-zone–year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Relative earnings and family beliefs

Additional stylized facts can be drawn from the *Family and Changing Gender Roles IV* module of the ISSP survey described in ISSP Research Group (2016). Respondents report their earnings relative to their spouse on a seven-point scale ranging from "My spouse/partner has no income" to "I have no income." Furthermore, they express agreement with a series of statements on family norms, such as "A preschool child is likely to suffer when the mother works" and "Family life suffers when a woman has a full-time job."[8]

Restricting the sample to U.S. couples and excluding the extreme tails of the earnings distribution,[9] I estimate the following specification:

$$\text{Belief}_i = \alpha + \sum_k \beta_k \, \mathbb{I}(\text{RelEarn}_i = k) + \mu X_i + \varepsilon_i, \tag{16}$$

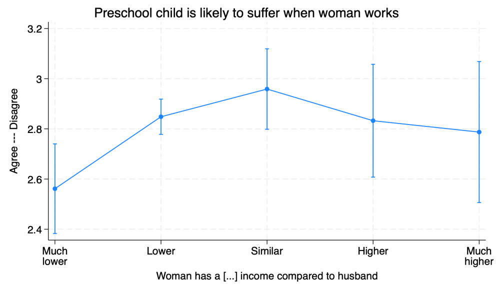
where the dependent variable is the agreement score for a given family-norm statement (higher score indicates disagreement); $\mathbb{I}(\text{RelEarn}_i = k)$ is a vector of dummy variables indicating the wife's earnings position relative to her husband; and X_i controls for respondent sex, both spouses' ages and education levels, and marriage duration. Standard errors are clustered by respondent and spouse education. Appendix Figure A1 presents the resulting marginal-effects plots.

Across statements, a distinct non-monotonicity emerges. For items stressing the disadvantages of maternal employment or endorsing traditional spousal roles (Panels a–d), the disagreement score follows an inverse-U pattern. Respondents move from agreement (low score) when the wife earns far less, to strong disagreement (high score) near earnings parity, and then revert back toward agreement once the wife clearly out-earns the husband. This reversion suggests that couples in non-traditional earnings arrangements may ostensibly adopt more traditional attitudes to compensate for the deviation from the norm.

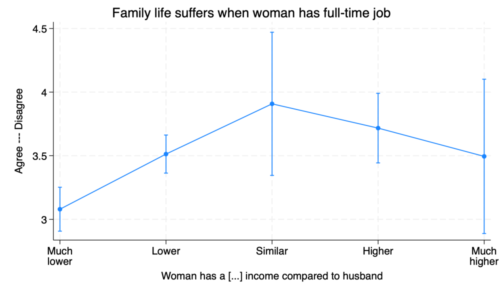
By contrast, attitudes toward maternal employment when children are older (Panel f) trace a U-shape in the disagreement score: couples at parity are the most likely to agree that women should work, while those with unequal earnings are more likely to disagree. Responses regarding maternal employment with preschool-age children (Panel e) display a similar but weaker downward trend, indicating less sensitivity to relative earnings for this specific issue.

[8]Two additional items—"Being a housewife is as fulfilling as working for pay" and "Both partners should contribute to household income"—were dropped because their directional interpretation regarding traditionalism is ambiguous.

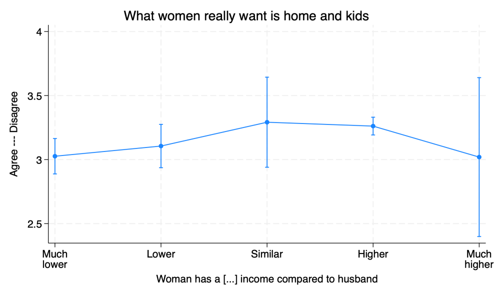
[9]Including extreme points strengthens the observed dynamics but may confound the results with involuntary unemployment effects.



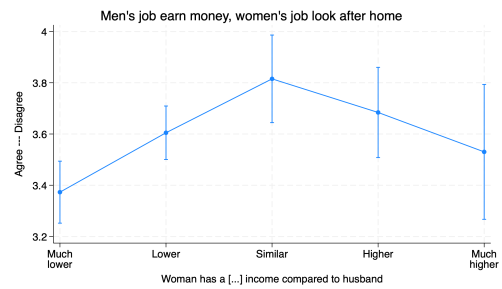
"Preschool child suffers if mother works"



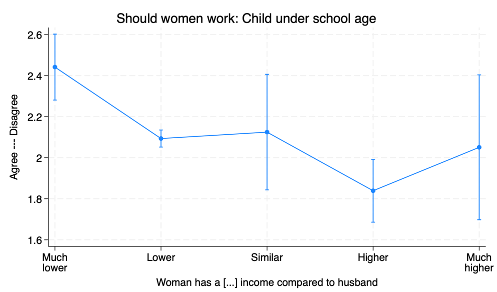
"Family life suffers if woman works full-time"



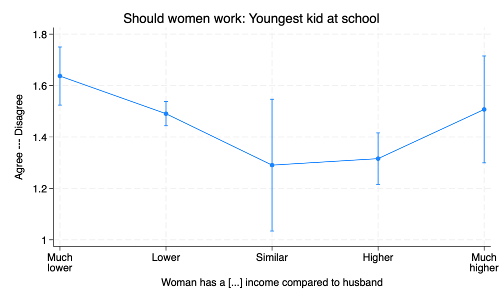
"What women really want is home/kids"



"Man's job is to earn money, woman's is home"



"Should women work: Child under school age"



"Should women work: Youngest kid at school"

Figure A1: Marginal effects of the wife's relative earnings on agreement with family-norm statements. The y-axis represents the disagreement score (higher = less traditional). The x-axis represents the wife's contribution to household income, ranging from much less than the husband (left) to much more (right).

C Solution of the semi-cooperative model

The Lagrangian of the second stage problem is:

$$L_i = \delta_i \log\bigl(c\bigr) + \mu_i \log\bigl(1 - (t_i + \tilde{t})\, n - h_i\bigr) + \gamma_i \log\bigl(n\,(1+t_i)^{\alpha}\,(1+t_{-i})^{1-\alpha}\bigr) + \eta\, t_i + \sigma\, t_{-i}$$

where η and σ are the Kuhn-Tucker multipliers associated to the non-negativity constraints on childcare time. The optimal solution satisfies $\eta t_f = \sigma t_m = 0$. The Nash Equilibrium of the second stage of the game can be in four regions, namely: $(\eta > 0, \sigma > 0), (\eta = 0, \sigma = 0), (\eta > 0, \sigma = 0), (\eta = 0, \sigma > 0)$.

The Lagrangian for the maximization of the first stage problem is

$$\begin{aligned} L = \theta\big(\delta_f \log(c) + \mu_f \log\big(1 - (t_f + \tilde{t})\, n - h_f\big) + \gamma_f \log\big(n\,(1+t_f)^{\alpha}\,(1+t_m)^{1-\alpha}\big)\big) \\ +(1-\theta)\big(\delta_m \log(c) + \mu_m \log\big(1 - (t_m + \tilde{t}_m)\, n - h_m\big) \\ +\gamma_m \log\big(n\,(1+t_f)^{\alpha}\,(1+t_m)^{1-\alpha}\big)\big) + \tau\, h_f + \nu\, h_m, \end{aligned}$$

where τ and ν are the Kuhn-Tucker multipliers associated to the non-negativity constraints on working time. The optimal solution satisfies $\tau h_f = \nu h_m = 0$, with the exception of the case $h_f = h_m = 0$. We get three possible solutions for each of the four regions defined in the second stage, for a total of 12 possible time allocation regimes, listed below.

Regime 1a: $\eta > 0, \sigma > 0, \tau = 0, \nu = 0$

$$h_m = \frac{\begin{aligned} &\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + (1-\theta)w_f + (\theta-1)\phi\Big)\,\mu_m \\ &+ \left(\theta n w_m \tilde{t}_m - \theta w_m\right)\mu_f \\ &+ \Big((1-\theta)nw_m\tilde{t}_m + (\theta-1)w_m\Big)\,\delta_m \\ &+ \left(\theta n w_m \tilde{t}_m - \theta w_m\right)\delta_f \end{aligned}}{(\theta-1)w_m\,\mu_m - \theta w_m\,\mu_f + (\theta-1)w_m\,\delta_m - \theta w_m\,\delta_f}$$

$$h_f = -\frac{\begin{aligned} &\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + (1-\theta)w_f + (\theta-1)\phi\Big)\,\mu_m \\ &+ \left(\theta n w_m \tilde{t}_m - \theta w_m\right)\mu_f \\ &+ \Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + (1-\theta)w_f + (\theta-1)\phi\Big)\,\delta_m \\ &+ \Big((\theta\phi n - \theta n w_f)\tilde{t}_f + \theta w_f - \theta\phi\Big)\,\delta_f \end{aligned}}{\begin{aligned} &\Big((\theta-1)w_f + (1-\theta)\phi\Big)\mu_m + (\theta\phi - \theta w_f)\mu_f \\ &+ \Big((\theta-1)w_f + (1-\theta)\phi\Big)\delta_m + (\theta\phi - \theta w_f)\delta_f \end{aligned}}$$

$$t_m = 0$$

$$t_f = 0$$

Regime 1b: $\eta > 0, \sigma > 0, \tau = 0, \nu > 0$

$$h_m = 0$$

$$h_f = \frac{\big((\theta-1)n\tilde{t}_f - \theta + 1\big)\delta_m + \big(\theta - \theta n\tilde{t}_f\big)\,\delta_f}{\theta\,\mu_f + (1-\theta)\delta_m + \theta\delta_f}$$

$$t_m = 0$$

$$t_f = 0$$

Regime 1c: $\eta > 0, \sigma > 0, \tau > 0, \nu = 0$

$$h_m = -\,\frac{\big((\theta-1)n\tilde{t}_m - \theta + 1\big)\,\delta_m + \big(\theta - \theta n\tilde{t}_m\big)\,\delta_f}{(\theta-1)\mu_m + (\theta-1)\delta_m - \theta\delta_f}$$

$$h_f = 0$$

$$t_m = 0$$

$$t_f = 0$$

Regime 2a: $\eta = 0, \sigma = 0, \tau = 0, \nu = 0$

$$h_m = \frac{\begin{aligned}&\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + \Big((1-\theta)n - \theta + 1\Big)w_f + (\theta-1)\phi n + (\theta-1)\phi\Big)\,\mu_m\\ &+\Big(\theta n w_m\tilde{t}_m + (-\theta n - \theta)w_m\Big)\,\mu_f\\ &+\Big((1-\theta)nw_m\tilde{t}_m + \big((\theta-1)n + \theta - 1\big)w_m\Big)\,\delta_m\\ &+\Big(\theta n w_m\tilde{t}_m + (-\theta n - \theta)w_m\Big)\,\delta_f\\ &+\Big((\alpha - \alpha\theta)nw_m\tilde{t}_m + \big((1-\alpha)\theta + \alpha - 1\big)nw_f\tilde{t}_f + \big((\alpha-1)\theta - \alpha + 1\big)\phi n\tilde{t}_f\\ &\quad + (\alpha\theta - \alpha)nw_m + \big(((\alpha-1)\theta - \alpha + 1)n + (\alpha-1)\theta - \alpha + 1\big)w_f\\ &\quad + \big((1-\alpha)\theta + \alpha - 1\big)\phi n + \big((1-\alpha)\theta + \alpha - 1\big)\phi\Big)\gamma_m\\ &+\Big(\alpha\theta n w_m\tilde{t}_m + \big((\alpha-1)\theta n w_f + (1-\alpha)\theta\phi n\big)\tilde{t}_f\\ &\quad + (-\alpha\theta n - \alpha\theta)w_m + \big((1-\alpha)\theta n + (1-\alpha)\theta\big)w_f + (\alpha-1)\theta\phi n + (\alpha-1)\theta\phi\Big)\gamma_f\end{aligned}}{\begin{aligned}&(\theta-1)w_m\,\mu_m - \theta w_m\,\mu_f + (\theta-1)w_m\,\delta_m - \theta w_m\,\delta_f\\ &\qquad + (\theta-1)w_m\,\gamma_m - \theta w_m\,\gamma_f\end{aligned}}$$

$$h_f = -\frac{\begin{aligned}&\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + \big((1-\theta)n - \theta + 1\big)w_f + (\theta-1)\phi n + (\theta-1)\phi\Big)\,\mu_m \\ &+ \Big(\theta n w_m \tilde{t}_m + (-\theta n - \theta)w_m\Big)\,\mu_f \\ &+ \Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + \big((1-\theta)n - \theta + 1\big)w_f + (\theta-1)\phi n + (\theta-1)\phi\Big)\,\delta_m \\ &+ \Big((\theta\phi n - \theta n w_f)\tilde{t}_f + (\theta n + \theta)w_f - \theta\phi n - \theta\phi\Big)\,\delta_f \\ &+ \Big((\alpha - \alpha\theta)nw_m\tilde{t}_m + \big((1-\alpha)\theta + \alpha - 1\big)nw_f\tilde{t}_f + \big((\alpha-1)\theta - \alpha + 1\big)\phi n\tilde{t}_f \\ &\quad + (\alpha\theta - \alpha)nw_m + \big(((\alpha-1)\theta - \alpha + 1)n + (\alpha-1)\theta - \alpha + 1\big)w_f \\ &\quad + \big((1-\alpha)\theta + \alpha - 1\big)\phi n + \big((1-\alpha)\theta + \alpha - 1\big)\phi\Big)\gamma_m \\ &+ \Big(\alpha\theta n w_m \tilde{t}_m + \big((\alpha-1)\theta n w_f + (1-\alpha)\theta\phi n\big)\tilde{t}_f \\ &\quad + (-\alpha\theta n - \alpha\theta)w_m + \big((1-\alpha)\theta n + (1-\alpha)\theta\big)w_f + (\alpha-1)\theta\phi n + (\alpha-1)\theta\phi\Big)\gamma_f\end{aligned}}{\begin{aligned}&\Big((\theta-1)w_f + (1-\theta)\phi\Big)\mu_m + (\theta\phi - \theta w_f)\mu_f \\ &+ \Big((\theta-1)w_f + (1-\theta)\phi\Big)\delta_m + (\theta\phi - \theta w_f)\delta_f \\ &+ \Big((\theta-1)w_f + (1-\theta)\phi\Big)\gamma_m + (\theta\phi - \theta w_f)\gamma_f\end{aligned}}$$

$$
t_m = -\frac{\begin{aligned}
&(\theta-1)nw_m\,\mu_m^2\\
&\quad+\Bigl(-\theta nw_m\,\mu_f+(\theta-1)nw_m\,\delta_m-\theta nw_m\,\delta_f\\
&\qquad+\Bigl(((1-\alpha)\theta+\alpha-1)nw_m\tilde{t}_m+((1-\alpha)\theta+\alpha-1)nw_f\tilde{t}_f\\
&\qquad\quad+\bigl((\alpha-1)\theta-\alpha+1\bigr)\phi n\tilde{t}_f+\bigl((\theta-1)n+(\alpha-1)\theta-\alpha+1\bigr)w_m\\
&\qquad\quad+\bigl(((\alpha-1)\theta-\alpha+1)n+(\alpha-1)\theta-\alpha+1\bigr)w_f\\
&\qquad\quad+\bigl((1-\alpha)\theta+\alpha-1\bigr)\phi n+\bigl((1-\alpha)\theta+\alpha-1\bigr)\phi\Bigr)\gamma_m-\theta nw_m\,\gamma_f\Bigr)\mu_m\\
&\quad+\alpha-1)\theta nw_m\,\gamma_m\,\mu_f\\
&\quad+\bigl((1-\alpha)\theta+\alpha-1\bigr)nw_m\,\gamma_m\,\delta_m+(\alpha-1)\theta nw_m\,\gamma_m\,\delta_f\\
&\quad+\Bigl(((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1)nw_m\tilde{t}_m\\
&\qquad+((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1)nw_f\tilde{t}_f\\
&\qquad+\bigl((-\alpha^2+2\alpha-1)\theta+\alpha^2-2\alpha+1\bigr)\phi n\tilde{t}_f\\
&\qquad+\bigl(((\alpha-\alpha^2)\theta+\alpha^2-\alpha)n+(-\alpha^2+2\alpha-1)\theta+\alpha^2-2\alpha+1\bigr)w_m\\
&\qquad+\bigl((-\alpha^2+2\alpha-1)\theta n+(-\alpha^2+2\alpha-1)\theta\bigr)w_f\\
&\qquad+\bigl((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\bigr)\phi n+\bigl((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\bigr)\phi\Bigr)\gamma_m^2\\
&\quad+\Bigl((-\alpha^2+2\alpha-1)\theta nw_m\tilde{t}_m+\bigl((-\alpha^2+2\alpha-1)\theta nw_f+(\alpha^2-2\alpha+1)\theta\phi n\bigr)\tilde{t}_f\\
&\qquad+\bigl((\alpha^2-\alpha)\theta n+(\alpha^2-2\alpha+1)\theta\bigr)w_m+\bigl((\alpha^2-2\alpha+1)\theta n+(\alpha^2-2\alpha+1)\theta\bigr)w_f\\
&\qquad+(-\alpha^2+2\alpha-1)\theta\phi n+(-\alpha^2+2\alpha-1)\theta\phi\Bigr)\gamma_f\gamma_m
\end{aligned}}{\begin{aligned}
&(\theta-1)nw_m\,\mu_m^2\\
&\quad+\Bigl(-\theta nw_m\,\mu_f+(\theta-1)nw_m\,\delta_m-\theta nw_m\,\delta_f\\
&\qquad+\bigl((2-2\alpha)\theta+2\alpha-2\bigr)nw_m\,\gamma_m-\theta nw_m\,\gamma_f\Bigr)\mu_m\\
&\quad+\alpha-1)\theta nw_m\,\gamma_m\,\mu_f\\
&\quad+\bigl((1-\alpha)\theta+\alpha-1\bigr)nw_m\,\gamma_m\,\delta_m+(\alpha-1)\theta nw_m\,\gamma_m\,\delta_f\\
&\quad+\bigl((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\bigr)nw_m\,\gamma_m^2\\
&\quad+\bigl(-\alpha^2+2\alpha-1\bigr)\theta nw_m\,\gamma_f\gamma_m
\end{aligned}}
$$

$$t_f = -\frac{\begin{aligned}&\big((\theta-1)nw_f+(1-\theta)\phi n\big)\mu_f\,\mu_m+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\mu_m\\&+\big(\theta\phi n-\theta nw_f\big)\mu_f^2\\&+\big((\theta-1)nw_f+(1-\theta)\phi n\big)\delta_m+\big(\theta\phi n-\theta nw_f\big)\delta_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_m\\&+\Big(-\alpha\theta nw_m\tilde{t}_m+(\alpha\theta\phi n-\alpha\theta nw_f)\tilde{t}_f+\alpha\theta w_m+(\alpha\theta-\alpha\theta n)w_f+\alpha\theta\phi n-\alpha\theta\phi\Big)\gamma_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\delta_m+\big(\alpha\theta\phi n-\alpha\theta nw_f\big)\gamma_f\delta_f\\&+\Big((\alpha^2\theta-\alpha^2)nw_m\tilde{t}_m+\big((\alpha^2\theta-\alpha^2)nw_f\\&+(\alpha^2-\alpha^2\theta)\phi n\big)\tilde{t}_f+(\alpha^2-\alpha^2\theta)(w_m+w_f)+(\alpha^2\theta-\alpha^2)\phi\Big)\gamma_f\gamma_m\\&+\Big(-\alpha^2\theta nw_m\tilde{t}_m+(\alpha^2\theta\phi n-\alpha^2\theta nw_f)\tilde{t}_f+\alpha^2\theta(w_m+w_f)-\alpha^2\theta\phi\Big)\gamma_f^2\end{aligned}}{\begin{aligned}&\big((\theta-1)nw_f+(1-\theta)\phi n\big)\mu_f\,\mu_m+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\mu_m\\&+\big(\theta\phi n-\theta nw_f\big)\mu_f^2\\&+\big((\theta-1)nw_f+(1-\theta)\phi n\big)\delta_m+\big(\theta\phi n-\theta nw_f\big)\delta_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_m\\&+\big(2\alpha\theta\phi n-2\alpha\theta nw_f\big)\gamma_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\delta_m+\big(\alpha\theta\phi n-\alpha\theta nw_f\big)\gamma_f\delta_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\gamma_m+\big(\alpha\theta\phi n-\alpha\theta nw_f\big)\gamma_f^2\end{aligned}}$$

Regime 2b: $\eta = 0, \sigma = 0, \tau = 0, \nu > 0$

$$h_m = 0$$

$$h_f = \frac{\big((\theta-1)n\tilde{t}_f+(1-\theta)n-\theta+1\big)\,\delta_m+\big(-\theta n\tilde{t}_f+\theta n+\theta\big)\,\delta_f}{\theta\,\mu_f+(1-\theta)\delta_m+\theta\delta_f+(\alpha-\alpha\theta)\gamma_m+\alpha\theta\gamma_f}$$

$$t_m = -\frac{n\,\mu_m+\big((1-\alpha)n\tilde{t}_m+\alpha-1\big)\gamma_m}{n\,\mu_m+(1-\alpha)n\,\gamma_m}$$

$$
t_f = -\frac{\begin{aligned}&\theta n\,\mu_f^2\\ &+\Big((1-\theta)n\,\delta_m+\theta n\,\delta_f+(\alpha-\alpha\theta)n\,\gamma_m+(\alpha\theta n\tilde{t}_f+\alpha\theta n-\alpha\theta)\gamma_f\Big)\mu_f\\ &+\big(\alpha-\alpha\theta\big)n\,\gamma_f\,\delta_m+\alpha\theta n\,\gamma_f\,\delta_f\\ &+\Big((\alpha^2-\alpha^2\theta)n\tilde{t}_f+\alpha^2\theta-\alpha^2\Big)\gamma_f\gamma_m+\big(\alpha^2\theta n\tilde{t}_f-\alpha^2\theta\big)\gamma_f^2\end{aligned}}{\begin{aligned}&\theta n\,\mu_f^2\\ &+\Big((1-\theta)n\,\delta_m+\theta n\,\delta_f+(\alpha-\alpha\theta)n\,\gamma_m+2\alpha\theta n\,\gamma_f\Big)\mu_f\\ &+\big(\alpha-\alpha\theta\big)n\,\gamma_f\,\delta_m+\alpha\theta n\,\gamma_f\,\delta_f\\ &+\big(\alpha^2-\alpha^2\theta\big)n\,\gamma_f\gamma_m+\alpha^2\theta n\,\gamma_f^2\end{aligned}}
$$

Regime 2c: $\eta = 0, \sigma = 0, \tau > 0, \nu = 0$

$$
h_m = -\frac{\big((\theta-1)n\tilde{t}_m+(1-\theta)n-\theta+1\big)\,\delta_m+\big(-\theta n\tilde{t}_m+\theta n+\theta\big)\,\delta_f}{(\theta-1)\mu_m+(\theta-1)\delta_m-\theta\delta_f+((1-\alpha)\theta+\alpha-1)\gamma_m+(\alpha-1)\theta\gamma_f}
$$

$$
h_f = 0
$$

$$
t_m = -\frac{\begin{aligned}&(\theta-1)n\,\mu_m^2\\ &+\Big((\theta-1)n\,\delta_m-\theta n\,\delta_f+\big(((1-\alpha)\theta+\alpha-1)n\tilde{t}_m\\ &+((1-\alpha)\theta+\alpha-1)n+(\alpha-1)\theta-\alpha+1\big)\gamma_m\quad+(\alpha-1)\theta n\,\gamma_f\Big)\mu_m\\ &+\big((1-\alpha)\theta+\alpha-1\big)n\gamma_m\,\delta_m+(\alpha-1)\theta n\gamma_m\,\delta_f\\ &+\Big(((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1)n\tilde{t}_m+(-\alpha^2+2\alpha-1)\theta+\alpha^2-2\alpha+1\Big)\gamma_m^2\\ &+\Big((-\alpha^2+2\alpha-1)\theta n\tilde{t}_m+(\alpha^2-2\alpha+1)\theta\Big)\gamma_f\gamma_m\end{aligned}}{\begin{aligned}&(\theta-1)n\,\mu_m^2\\ &+\Big((\theta-1)n\,\delta_m-\theta n\,\delta_f+\big((2-2\alpha)\theta+2\alpha-2\big)n\gamma_m+(\alpha-1)\theta n\,\gamma_f\Big)\mu_m\\ &+\big((1-\alpha)\theta+\alpha-1\big)n\gamma_m\,\delta_m+(\alpha-1)\theta n\gamma_m\,\delta_f\\ &+\big((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\big)n\gamma_m^2+(-\alpha^2+2\alpha-1)\theta n\,\gamma_f\gamma_m\end{aligned}}
$$

$$
t_f = -\frac{n\,\mu_f+(\alpha n\tilde{t}_f-\alpha)\gamma_f}{n\,\mu_f+\alpha n\,\gamma_f}
$$

Regime 3a: $\eta > 0, \sigma = 0, \tau = 0, \nu = 0$

$$h_m = \frac{\begin{aligned}&\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + (1-\theta)w_f + (\theta-1)\phi\Big)\,\mu_m \\ &+ \Big(\theta n w_m \tilde{t}_m + (-\theta n - \theta)w_m\Big)\,\mu_f \\ &+ \Big((1-\theta)nw_m\tilde{t}_m + \big((\theta-1)n + \theta - 1\big)w_m\Big)\,\delta_m \\ &+ \Big(\theta n w_m \tilde{t}_m + (-\theta n - \theta)w_m\Big)\,\delta_f \\ &+ \Big(\big((1-\alpha)\theta + \alpha - 1\big)nw_f + \big((\alpha-1)\theta - \alpha + 1\big)\phi n\Big)\tilde{t}_f \\ &+ \big((\alpha-1)\theta - \alpha + 1\big)w_f + \big((1-\alpha)\theta + \alpha - 1\big)\phi\Big)\,\gamma_m \\ &+ \Big(\big((\alpha-1)\theta n w_f + (1-\alpha)\theta\phi n\big)\tilde{t}_f + (1-\alpha)\theta w_f + (\alpha-1)\theta\phi\Big)\,\gamma_f\end{aligned}}{\begin{aligned}&(\theta-1)w_m\,\mu_m - \theta w_m\,\mu_f \\ &+ \big((\theta-1)w_m - \theta w_m\big)\delta_m - \theta w_m\,\delta_f \\ &+ \big((1-\alpha)\theta + \alpha - 1\big)w_m\,\gamma_m + (\alpha-1)\theta w_m\,\gamma_f\end{aligned}}$$

$$h_f = -\frac{\begin{aligned}&\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + (1-\theta)w_f + (\theta-1)\phi\Big)\,\mu_m \\ &+ \Big(\theta n w_m \tilde{t}_m + (-\theta n - \theta)w_m\Big)\,\mu_f \\ &+ \Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + (1-\theta)w_f + (\theta-1)\phi\Big)\,\delta_m \\ &+ \Big((\theta\phi n - \theta n w_f)\tilde{t}_f + \theta w_f - \theta\phi\Big)\,\delta_f \\ &+ \Big(\big((1-\alpha)\theta + \alpha - 1\big)nw_f + \big((\alpha-1)\theta - \alpha + 1\big)\phi n\Big)\tilde{t}_f + \big((\alpha-1)\theta - \alpha + 1\big)w_f + \big((1-\alpha)\theta \\ &+ \alpha - 1\big)\phi\Big)\,\gamma_m + \Big(\big((\alpha-1)\theta n w_f + (1-\alpha)\theta\phi n\big)\tilde{t}_f + (1-\alpha)\theta w_f + (\alpha-1)\theta\phi\Big)\,\gamma_f\end{aligned}}{\begin{aligned}&\Big((\theta-1)w_f + (1-\theta)\phi\Big)\mu_m + (\theta\phi - \theta w_f)\mu_f \\ &+ \Big((\theta-1)w_f + (1-\theta)\phi\Big)\delta_m + (\theta\phi - \theta w_f)\delta_f \\ &+ \big((1-\alpha)\theta + \alpha - 1\big)w_f + \big((\alpha-1)\theta - \alpha + 1\big)\phi\Big)\gamma_m \\ &+ \big((\alpha-1)\theta w_f + (1-\alpha)\theta\phi\big)\gamma_f\end{aligned}}$$

$$t_m = -\frac{\begin{aligned}&(\theta-1)nw_m\,\mu_m^2\\&+\Big(-\theta nw_m\,\mu_f+(\theta-1)nw_m\,\delta_m-\theta nw_m\,\delta_f\\&\quad+\Big(((1-\alpha)\theta+\alpha-1)nw_m\tilde{t}_m+((1-\alpha)\theta+\alpha-1)nw_f\tilde{t}_f\\&\qquad+\big((\alpha-1)\theta-\alpha+1\big)\phi n\tilde{t}_f+\big(((1-\theta)+\alpha-1)n+(\alpha-1)\theta-\alpha+1\big)w_m\\&\qquad+\big((\alpha-1)\theta-\alpha+1\big)w_f+\big((1-\alpha)\theta+\alpha-1\big)\phi\Big)\gamma_m+(\alpha-1)\theta nw_m\,\gamma_f\Big)\mu_m\\&+\big(\alpha-1\big)\theta nw_m\,\gamma_m\,\mu_f\\&+\big((1-\alpha)\theta+\alpha-1\big)nw_m\,\gamma_m\,\delta_m+(\alpha-1)\theta nw_m\,\gamma_m\,\delta_f\\&+\Big(((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1)nw_m\tilde{t}_m\\&\quad+((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1)nw_f\tilde{t}_f\\&\quad+\big((-\alpha^2+2\alpha-1)\theta+\alpha^2-2\alpha+1\big)\phi n\tilde{t}_f\\&\quad+\big((-\alpha^2+2\alpha-1)\theta+\alpha^2-2\alpha+1\big)w_m\\&\quad+\big((-\alpha^2+2\alpha-1)\theta+\alpha^2-2\alpha+1\big)w_f\\&\quad+\big((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\big)\phi\Big)\gamma_m^2\\&+\Big((-\alpha^2+2\alpha-1)\theta nw_m\tilde{t}_m+\big((-\alpha^2+2\alpha-1)\theta nw_f+(\alpha^2-2\alpha+1)\theta\phi n\big)\tilde{t}_f\\&\quad+\big((\alpha^2-2\alpha+1)\theta+\alpha^2-2\alpha+1\big)w_m\\&\quad+\big((\alpha^2-2\alpha+1)\theta+\alpha^2-2\alpha+1\big)w_f+(-\alpha^2+2\alpha-1)\theta\phi\Big)\gamma_f\gamma_m\end{aligned}}{\begin{aligned}&(\theta-1)nw_m\,\mu_m^2\\&+\Big(-\theta nw_m\,\mu_f+(\theta-1)nw_m\,\delta_m-\theta nw_m\,\delta_f\\&\quad+\big((2-2\alpha)\theta+2\alpha-2\big)nw_m\,\gamma_m+(\alpha-1)\theta nw_m\,\gamma_f\Big)\mu_m\\&+\big(\alpha-1\big)\theta nw_m\,\gamma_m\,\mu_f\\&+\big((1-\alpha)\theta+\alpha-1\big)nw_m\,\gamma_m\,\delta_m+(\alpha-1)\theta nw_m\,\gamma_m\,\delta_f\\&+\big((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\big)nw_m\,\gamma_m^2+(-\alpha^2+2\alpha-1)\theta nw_m\,\gamma_f\gamma_m\end{aligned}}$$

$$t_f = 0$$

Regime 3b: $\eta > 0, \sigma = 0, \tau = 0, \nu > 0$

$$h_m = 0$$

$$h_f = \frac{\big((\theta-1)n\tilde{t}_f - \theta + 1\big)\,\delta_m + \big(\theta - \theta n\tilde{t}_f\big)\,\delta_f}{\theta\,\mu_f + (1-\theta)\delta_m + \theta\delta_f}$$

$$t_m = -\frac{n\,\mu_m + \big((1-\alpha)n\tilde{t}_m + \alpha - 1\big)\gamma_m}{n\,\mu_m + (1-\alpha)n\,\gamma_m}$$

Regime 3c: $\eta > 0, \sigma = 0, \tau > 0, \nu = 0$

$$h_m = -\frac{\big((\theta-1)n\tilde{t}_m + (1-\theta)n - \theta + 1\big)\,\delta_m + \big(-\theta n\tilde{t}_m + \theta n + \theta\big)\,\delta_f}{(\theta-1)\mu_m + (\theta-1)\delta_m - \theta\delta_f + \big((1-\alpha)\theta + \alpha - 1\big)\gamma_m + (\alpha-1)\theta\gamma_f}$$

$$h_f = 0$$

$$t_m = -\frac{\begin{aligned}&(\theta-1)n\,\mu_m^2\\&+\Big((\theta-1)n\,\delta_m - \theta n\,\delta_f + \big((1-\alpha)\theta+\alpha-1\big)n\tilde{t}_m + \big((1-\alpha)\theta+\alpha-1\big)n\\&\quad+(\alpha-1)\theta-\alpha+1\Big)\gamma_m + (\alpha-1)\theta n\,\gamma_f\Big)\mu_m\\&+\big((1-\alpha)\theta+\alpha-1\big)n\gamma_m\,\delta_m + (\alpha-1)\theta n\gamma_m\,\delta_f\\&+\big((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\big)n\tilde{t}_m + \big(-\alpha^2+2\alpha-1\big)\theta+\alpha^2-2\alpha+1\Big)\gamma_m^2\\&+\Big((-\alpha^2+2\alpha-1)\theta n\tilde{t}_m + (\alpha^2-2\alpha+1)\theta\Big)\gamma_f\gamma_m\end{aligned}}{\begin{aligned}&(\theta-1)n\,\mu_m^2\\&+\Big((\theta-1)n\,\delta_m - \theta n\,\delta_f + \big((2-2\alpha)\theta+2\alpha-2\big)n\gamma_m + (\alpha-1)\theta n\,\gamma_f\Big)\mu_m\\&+\big((1-\alpha)\theta+\alpha-1\big)n\gamma_m\,\delta_m + (\alpha-1)\theta n\gamma_m\,\delta_f\\&+\big((\alpha^2-2\alpha+1)\theta-\alpha^2+2\alpha-1\big)n\gamma_m^2 + \big(-\alpha^2+2\alpha-1\big)\theta n\,\gamma_f\gamma_m\end{aligned}}$$

$$t_f = 0$$

Regime 4a: $\eta = 0, \sigma > 0, \tau = 0, \nu = 0$

$$h_m = \frac{\begin{aligned}&\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + \Big((1-\theta)n - \theta + 1\Big)w_f + (\theta-1)\phi n + (\theta-1)\phi\Big)\,\mu_m \\ &+ \left(\theta n w_m \tilde{t}_m - \theta w_m\right)\mu_f \\ &+ \Big((1-\theta)nw_m\tilde{t}_m + (\theta-1)w_m\Big)\,\delta_m \\ &+ \left(\theta n w_m \tilde{t}_m - \theta w_m\right)\delta_f \\ &+ \Big((\alpha-\alpha\theta)nw_m\tilde{t}_m + (\alpha\theta-\alpha)w_m\Big)\,\gamma_m \\ &+ \Big(\alpha\theta n w_m\tilde{t}_m - \alpha\theta w_m\Big)\,\gamma_f\end{aligned}}{\begin{aligned}&(\theta-1)w_m\,\mu_m - \theta w_m\,\mu_f + (\theta-1)w_m\,\delta_m - \theta w_m\,\delta_f \\ &+(\alpha\theta-\alpha)w_m\,\gamma_m - \alpha\theta w_m\,\gamma_f\end{aligned}}$$

$$h_f = -\frac{\begin{aligned}&\Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + \Big((1-\theta)n - \theta + 1\Big)w_f + (\theta-1)\phi n + (\theta-1)\phi\Big)\,\mu_m \\ &+ \left(\theta n w_m \tilde{t}_m - \theta w_m\right)\mu_f \\ &+ \Big((\theta-1)nw_f + (1-\theta)\phi n\Big)\tilde{t}_f + \Big((1-\theta)n - \theta + 1\Big)w_f + (\theta-1)\phi n + (\theta-1)\phi\Big)\,\delta_m \\ &+ \Big((\theta\phi n - \theta n w_f)\tilde{t}_f + (\theta n + \theta)w_f - \theta\phi n - \theta\phi\Big)\,\delta_f \\ &+ \Big((\alpha-\alpha\theta)nw_m\tilde{t}_m + (\alpha\theta-\alpha)w_m\Big)\,\gamma_m \\ &+ \Big(\alpha\theta n w_m\tilde{t}_m - \alpha\theta w_m\Big)\,\gamma_f\end{aligned}}{\begin{aligned}&\Big((\theta-1)w_f + (1-\theta)\phi\Big)\mu_m + (\theta\phi - \theta w_f)\mu_f \\ &+ \Big((\theta-1)w_f + (1-\theta)\phi\Big)\delta_m + (\theta\phi - \theta w_f)\delta_f \\ &+ \Big((\alpha\theta-\alpha)w_f + (\alpha-\alpha\theta)\phi\Big)\gamma_m + (\alpha\theta\phi - \alpha\theta w_f)\gamma_f\end{aligned}}$$

$$t_m = 0$$

$$t_f = -\frac{\begin{aligned}&\Big((\theta-1)nw_f+(1-\theta)\phi n\Big)\mu_f+\Big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\Big)\gamma_f\Big)\mu_m\\&+\big(\theta\phi n-\theta nw_f\big)\mu_f^2\\&+\Big((\theta-1)nw_f+(1-\theta)\phi n\Big)\delta_m+\big(\theta\phi n-\theta nw_f\big)\delta_f\\&+\Big((\theta-1)nw_f+(1-\theta)\phi n\Big)\gamma_m\\&+\Big(-\alpha\theta nw_m\tilde{t}_m+(\alpha\theta\phi n-\alpha\theta nw_f)\tilde{t}_f+(\alpha\theta n+\alpha\theta)w_m+(\alpha\theta-\theta n)w_f+\theta\phi n-\alpha\theta\phi\Big)\gamma_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\delta_m+\big(\alpha\theta\phi n-\alpha\theta nw_f\big)\gamma_f\delta_f\\&+\Big((\alpha^2\theta-\alpha^2)nw_m\tilde{t}_m+\big((\alpha^2\theta-\alpha^2)nw_f\\&+(\alpha^2-\alpha^2\theta)\phi n\big)\tilde{t}_f+(\alpha^2-\alpha^2\theta)(w_m+w_f)+(\alpha^2\theta-\alpha^2)\phi\Big)\gamma_f\gamma_m\\&+\Big(-\alpha^2\theta nw_m\tilde{t}_m+(\alpha^2\theta\phi n-\alpha^2\theta nw_f)\tilde{t}_f+\alpha^2\theta(w_m+w_f)-\alpha^2\theta\phi\Big)\gamma_f^2\end{aligned}}{\begin{aligned}&\Big((\theta-1)nw_f+(1-\theta)\phi n\Big)\mu_f+\Big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\Big)\gamma_f\Big)\mu_m\\&+\big(\theta\phi n-\theta nw_f\big)\mu_f^2\\&+\Big((\theta-1)nw_f+(1-\theta)\phi n\Big)\delta_m+\big(\theta\phi n-\theta nw_f\big)\delta_f\\&+\Big((\theta-1)nw_f+(1-\theta)\phi n\Big)\gamma_m\\&+\big((-\alpha-1)\theta nw_f+(\alpha+1)\theta\phi n\big)\gamma_f\,\mu_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\delta_m+\big(\alpha\theta\phi n-\alpha\theta nw_f\big)\gamma_f\delta_f\\&+\big((\alpha\theta-\alpha)nw_f+(\alpha-\alpha\theta)\phi n\big)\gamma_f\gamma_m+\big(\alpha\theta\phi n-\alpha\theta nw_f\big)\gamma_f^2\end{aligned}}$$

Regime 4b: $\eta = 0, \sigma > 0, \tau = 0, \nu > 0$

$$h_m = 0$$

$$h_f = \frac{\left((\theta-1)n\tilde{t}_f+(1-\theta)n-\theta+1\right)\delta_m+\left(-\theta n\tilde{t}_f+\theta n+\theta\right)\delta_f}{\theta\,\mu_f+(1-\theta)\delta_m+\theta\delta_f+(\alpha-\alpha\theta)\gamma_m+\alpha\theta\gamma_f}$$

$$t_m = 0$$

$$
t_f = -\frac{\begin{aligned}&\theta n\,\mu_f^2 + \Big((1-\theta)n\,\delta_m + \theta n\,\delta_f + (\alpha-\alpha\theta)n\,\gamma_m + (\alpha\theta n\tilde{t}_f + \alpha\theta n - \alpha\theta)\gamma_f\Big)\mu_f \\ &+ \big(\alpha-\alpha\theta\big)n\,\gamma_f\,\delta_m + \alpha\theta n\,\gamma_f\,\delta_f \\ &+ \Big((\alpha^2-\alpha^2\theta)n\tilde{t}_f + \alpha^2\theta - \alpha^2\Big)\gamma_f\gamma_m + \big(\alpha^2\theta n\tilde{t}_f - \alpha^2\theta\big)\gamma_f^2\end{aligned}}{\begin{aligned}&\theta n\,\mu_f^2 + \Big((1-\theta)n\,\delta_m + \theta n\,\delta_f + (\alpha-\alpha\theta)n\,\gamma_m + 2\alpha\theta n\,\gamma_f\Big)\mu_f \\ &+ \big(\alpha-\alpha\theta\big)n\,\gamma_f\,\delta_m + \alpha\theta n\,\gamma_f\,\delta_f \\ &+ \big(\alpha^2-\alpha^2\theta\big)n\,\gamma_f\gamma_m + \alpha^2\theta n\,\gamma_f^2\end{aligned}}
$$

Regime 4c: $\eta = 0, \sigma > 0, \tau > 0, \nu = 0$

$$
h_m = -\frac{\big((\theta-1)n\tilde{t}_m - \theta + 1\big)\,\delta_m + \big(\theta - \theta n\tilde{t}_m\big)\,\delta_f}{(\theta-1)\mu_m + (\theta-1)\delta_m - \theta\delta_f}
$$

$$
h_f = 0
$$

$$
t_m = 0
$$

$$
t_f = -\frac{n\,\mu_f + (\alpha n\tilde{t}_f - \alpha)\gamma_f}{n\,\mu_f + \alpha n\,\gamma_f}
$$

C.1 Proof of Proposition 1

Stage 2. Fix (h_f, h_m). Player i solves a strictly concave problem in t_i:

$$
v_i(t_i; t_{-i}, h_f, h_m) = \log(c) + \mu_i \log\big(1 - h_i - (t_i + \tilde{t}_i)n\big) + \gamma_i \log\big((1+t_f)^\alpha (1+t_m)^{1-\alpha} n\big),
$$

over a nonempty, convex, compact interval for t_i (A3). The second derivative in own strategy,

$$
\frac{\partial^2 v_i}{\partial t_i^2} = -\frac{\mu_i n^2}{\big(1 - h_i - (t_i + \tilde{t}_i)n\big)^2} - \frac{\gamma_i}{(1+t_i)^2} \times \begin{cases}\alpha & \text{if } i = f,\\ 1-\alpha & \text{if } i = m,\end{cases}
$$

is strictly negative, so each payoff is strictly concave in own action. The Jacobian of first-order conditions has a symmetric part whose diagonal entries dominate the off-diagonal terms (the cross–partial from $\log q$), implying diagonal strict concavity (DSC). By Rosen's (1965) theorem, a unique Nash equilibrium (t_f^*, t_m^*) exists; by the implicit function theorem and DSC, (t_f^*, t_m^*) is continuous in (h_f, h_m).

Stage 1. Define $V(h_f, h_m) = \theta v_f\big(h_f, h_m, t^*(h)\big) + (1-\theta) v_m\big(h_f, h_m, t^*(h)\big)$. The feasible set for (h_f, h_m) is convex and compact (A3). The only non-affine term in (h_f, h_m) is $\log(c)$ with $c = \phi(s) w_f h_f + w_m h_m$. Because s is the exogenous potential wage share, $\phi(s)$ is independent of (h_f, h_m). Therefore, c is a linear combination of market hours, and the mapping $(h_f, h_m) \mapsto \log(c)$ is strictly concave on the feasible set. All remaining terms in V are concave in (h_f, h_m), and composition with the single-valued continuous correspondence $t^*(h)$ preserves concavity. Hence V is strictly concave and attains a unique maximizer (h_f^*, h_m^*). Combining with the unique (t_f^*, t_m^*) from Stage 2 yields a unique subgame-perfect equilibrium.

□